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Pathology Public Datasets for Artificial Intelligence: A Systematic Review

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Abstract
Histopathology plays a crucial role in the diagnosis of many diseases, especially cancers, for which the correct classification of tissue samples significantly influences treatments for patients. The growing use of artificial intelligence (AI) in digital pathology offers opportunities for improving diagnostic process speed, accuracy, and scalability. The availability of well-structured, annotated histopathology datasets is essential to this advancement. This study provides an extensive overview of publicly available datasets tailored specifically for histopathology-related AI and machine learning research. Our review yielded 151 datasets across tissue types and cancers (gastrointestinal, brain glioma, lung adenocarcinoma, and others). We also categorize the datasets in terms of the number of patients, organs, staining, magnification, scanner, size, collected method, year, and resolution. We analyze multiple key and popular datasets, including but not limited to CAMELYON, TUPAC, MIDOG, MoNuSeg, and BreakHis. We believe this review will help computational histopathology research by providing a comprehensive understanding of the available datasets, their structures, and their specific applications. Researchers can more effectively choose relevant datasets for creating AI models suited to certain tasks, such as cancer diagnosis, treatment response prediction, and tissue classification, by documenting and evaluating these resources. Within the field, standardizing images across these datasets can facilitate collaborations between the data-generating experts, pathologists, and AI model developers, as well as help in reproducibility, benchmark testing, and evaluation. Furthermore, by combining histopathological, radiological, and genomic data, for example, this evaluation will help identify gaps in the availability of current datasets. Another benefit is that it will help identify the need for additional diversified datasets that incorporate multimodal data. Closing these gaps will be essential to creating AI models that are applicable to different types of institutions and patient groups.

Keywords Computational pathology · Digital pathology · Datasets · Deep learning · Artificial intelligence · Data standardization · Segmentation · Classification · Whole slide imaging · Big data · Review

Introduction

Traditionally, pathologists' expertise has been crucial to the area of histopathology, which deals with the microscopic examination of tissue samples to identify and diagnose diseases like cancer. However, recent advances in machine learning and artificial intelligence (AI) have created opportunities for improving the overall efficiency of histopathology studies as well as improving diagnostic accuracy and lowering subjectivity. The integration of artificial intelligence into pathology holds immense promise for enhancing diagnostic accuracy, efficiency, and standardization. Developing robust, generalizable, and clinically relevant AI models specifically for histopathology has become a crucial field of study as the need for automated systems grows, particularly in the field of medical diagnostics [1]. The availability of high-quality and large-scale histopathology datasets is fundamental to these advances. These digital pathology image collections, which include diagnostic annotations, are crucial for AI model training and validation. Researchers can create robust models that can carry out tasks like cancer detection, tumor grading, and therapeutic response prediction by using well-curated datasets. Importantly, datasets that reflect variability in imaging techniques, staining processes, and patient demographics are critical for ensuring the generalizability of AI models across different clinical settings.

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Even with the increasing accessibility and public availability of datasets, there are still obstacles to overcome to fully utilize these resources [2]. The imaging modality, staining protocols, magnification levels, and tissue types may differ greatly between each dataset. Furthermore, certain tasks, such as segmentation [3, 4] or classification [5, 6], are better suited for particular datasets, whereas prediction or disease progression modeling [7] tasks are better suited for other datasets. Furthermore, problems including class imbalances, domain shifts, and noisy labeling make it more difficult to create AI models. With the advent of foundation models (FMs), it has become imperative that we need a large amount of data. Histopathology FMs such as Virchow [8], UNI [9], GIGAPATH [10] are trained on 100k WSIs, 100M tiles, and 171k WSIs, 1.4B tiles respectively. There are now more than 50 FMs in histopathology with datasets containing 3000 to 1.5 million WSIs; we refer to [11–13] for recent reviews on these models.

In addition to the increasing availability of pathology datasets, advancements in AI-driven pathology are also being shaped by improvements in computational infrastructure and deep learning techniques. The ability to train AI models on high-resolution whole-slide images (WSIs) has been made possible through advancements in cloud computing, graphical processing unit (GPU) acceleration, and optimized data pipelines. These developments have significantly reduced the computational bottlenecks associated with handling large-scale histopathology images, enabling more efficient model training and inference. Furthermore, recent innovations in self-supervised learning, contrastive learning, and transformer-based architectures have demonstrated promising results in learning from large, unannotated pathology datasets. As dataset sizes continue to grow, ensuring that AI algorithms can effectively leverage these datasets with minimal manual annotation will be a key factor in advancing digital pathology research. In this work, we provide a systematic review of pathology datasets that are used in developing and benchmarking AI models. By collectively summarizing, we provide a succinct overview of the available datasets in terms of their organs, staining, and detailed descriptions. To the best of our knowledge, there are no existing reviews of this nature. However, there are various pathology AI reviews and surveys [4, 14–18]. While these previous reviews focused on specific models, techniques, or narrow tasks like nuclei segmentation and image classification, our work is the first to categorize 151 datasets across the entire histopathology spectrum, providing a longitudinal timeline of data explosion in pathology. The digital transformation of pathology is not merely a technical shift but a clinical necessity driven by the rising global burden of cancer and a dwindling workforce of specialist pathologists. The clinical significance of the 151 datasets identified in this review lies in their potential to standardize diagnostic criteria and reduce the inherent subjectivity in manual tissue assessment. Traditionally, histopathology relies on the qualitative expertise of a pathologist to identify complex morphological patterns. However, inter-observer variability remains a challenge in critical tasks. For example, in tumor grading, datasets like DiagSet (prostate) and IDC Histology (breast) provide the massive scale of annotated examples needed to train AI in identifying subtle cellular changes, such as the Gleason grading system, which directly determines surgical and therapeutic paths. In metastasis detection, identifying micrometastases in sentinel lymph nodes is a labor-intensive task where human fatigue can lead to oversight. Foundational resources like CAMELYON16 and CAMELYON17 have enabled the development of "clinical-grade" algorithms that act as a second-look system, ensuring higher diagnostic sensitivity for oncology patients. Beyond initial diagnosis, the integration of AI-ready datasets is pivotal for precision oncology. Modern pathology now demands "predictive" insights that go beyond what is visible to the naked eye. In biomarker quantification, automating the scoring of biomarkers like HER2 (via the HER2-Warwick and HEROHE datasets) or PD-L1 (via the IMPRESS dataset) ensures that patients receive the most appropriate targeted therapies based on objective, reproducible measurements. Similarly, in treatment prediction, datasets such as Ovarian Bevacizumab Response and Post-NAT-BRCA are essential for training models to predict how a patient will respond to chemotherapy or immunotherapy, allowing clinicians to adjust treatment plans before adverse effects occur. Ultimately, the availability of these high-quality, annotated datasets is the "clinical fuel" for the next generation of diagnostic tools. By closing the gap between raw data and actionable clinical insights, these resources help ensure that AI models are generalizable across different institutions and diverse patient groups, ultimately improving healthcare outcomes worldwide. We believe this provides a comprehensive collection of AI-ready pathology datasets that can help researchers tailor machine learning and AI research pipelines. We discuss perspectives gained from the review of these datasets, along with challenges that still remain.

Method

We performed a comprehensive search using Google Scholar, PubMed, ScienceDirect, IEEE Xplore, and Scopus databases, based on a set of keywords including "histopathology dataset," "cancer histopathology dataset," "AI-based histopathology," "deep learning in pathology," and "AI for cancer detection." We searched on these databases until the end of September 2025. Initially, we identified a total of 2874 articles, which were filtered for duplicates, resulting in 353 unique references and datasets. From this pool, we applied

additional filters to exclude irrelevant studies based on three primary criteria:

1. The datasets were unrelated to histopathology.
2. The papers/datasets did not focus on image analysis tasks relevant to AI, such as classification, segmentation, or detection.
3. The datasets used in the papers were not publicly available or lacked sufficient annotations. This criterion is applied by carefully vetting the corresponding online repositories and links.

After applying these constraints, 151 papers remained. By narrowing our scope to focus on publicly available datasets relevant to AI, diagnosis, and treatment, we ensured that the 151 datasets selected for this review are of high quality and relevance to current trends in computational pathology research. These datasets serve as foundational resources for developing robust AI models, which can improve diagnostic accuracy and help standardize the assessment of histopathological images across various clinical settings. Two authors (ARC, SSD) independently surveyed the characteristics of the datasets that were found based on the eligibility criteria. The details recorded include direct link or digital object identifier (DOI), associated publication or citation, year, version, accessibility (fully public, controlled access), and content of the dataset including the number of WSIs, images, type (raw, processed), and other notable features. The third author (VBSP) independently verified all the resultant datasets as well as acted as a tiebreaker on exclusion criteria. We also further undertook a snowballing strategy—by exploring citations of papers that describe the particular dataset—to look for subsequent and related datasets. The final result of 151 datasets represents various tissue types from different organs, magnification factor, scanners, sizes, collected methods, year, number of patients, and resolution levels.

Systematic Review Results

Figure 1 shows the preferred reporting items for systematic reviews and meta-analyses (PRISMA) [19, 20] flow of the systematic review of pathology datasets and their corresponding publications. We note that the final collection of datasets was carefully identified along with a corresponding publication as per the dataset's metadata description for relevant citations. There are 151 datasets, and these are described next succinctly in terms of the organs, magnification, and the number of patients, samples, and whole-slide images (WSIs) they contain, along with their use cases and labels.

1. BreakHis [21]: The BreakHis dataset contains 7909 histopathological images of breast tumors from 82 patients, including benign (2480 images) and malignant (5429 images) cases. Captured at magnifications of 40×, 100×, 200×, and 400×, it supports breast cancer research by providing a detailed foundation for developing computer-aided diagnosis (CAD) systems. The dataset includes four benign tumor subtypes and four malignant tumor types, offering diverse diagnostic information to improve breast cancer detection and classification.
2. LC25000 [22]: LC25000 consists of 25,000 histopathological images of lung and colon tissues, divided into five classes with 5000 images each. These include adenocarcinoma and benign tissues from both organs. The images, initially captured from pathology slides, were augmented to expand the dataset. It is freely available for developing machine learning algorithms in cancer pathology.
3. Ovarian Bevacizumab Response [23]: This dataset includes 288 WSIs of epithelial ovarian cancer and peritoneal serous papillary carcinoma from 78 patients. The dataset provides both effective and resistant treatment outcomes, helping researchers develop methods to predict bevacizumab responses. It includes detailed clinical data, supporting personalized cancer treatment.
4. PLISM (PLISM-original, PLISM-WSI, PLISM-sm [24]): The PLISM dataset, created by researchers at the University of Tokyo, addresses the challenge of domain shifts in pathology analysis. It includes 91 high-resolution pathology images and two subsets: PLISM-sm, containing 60,000 smartphone image tiles, and PLISM-wsi, with 300,000 WSI tiles. Both subsets capture variations in tissue staining and imaging, enabling the development of adaptable AI models for pathology.
5. Chaoyang [25]: The Chaoyang dataset, part of a study on noise-robust learning methods, contains colon tissue slides processed into 512 × 512 pixel patches, labeled by multiple pathologists. It simulates real-world challenges in histopathology, with label discrepancies reflecting inter-observer variability. This dataset, along with synthetic noise datasets, aids in evaluating noise-robust classification models.
6. DiagSet [26]: DiagSet is a large dataset for prostate cancer detection, featuring over 2.6 million tissue patches categorized by Gleason grading. It includes fully annotated and binary-diagnosed scans, providing a robust resource for developing machine learning models for prostate cancer detection and classification. The dataset facilitates patch-level and scan-level prostate cancer research.
7. BCNB [27]: The BCNB dataset contains WSIs from 1058 early breast cancer patients, annotated by two

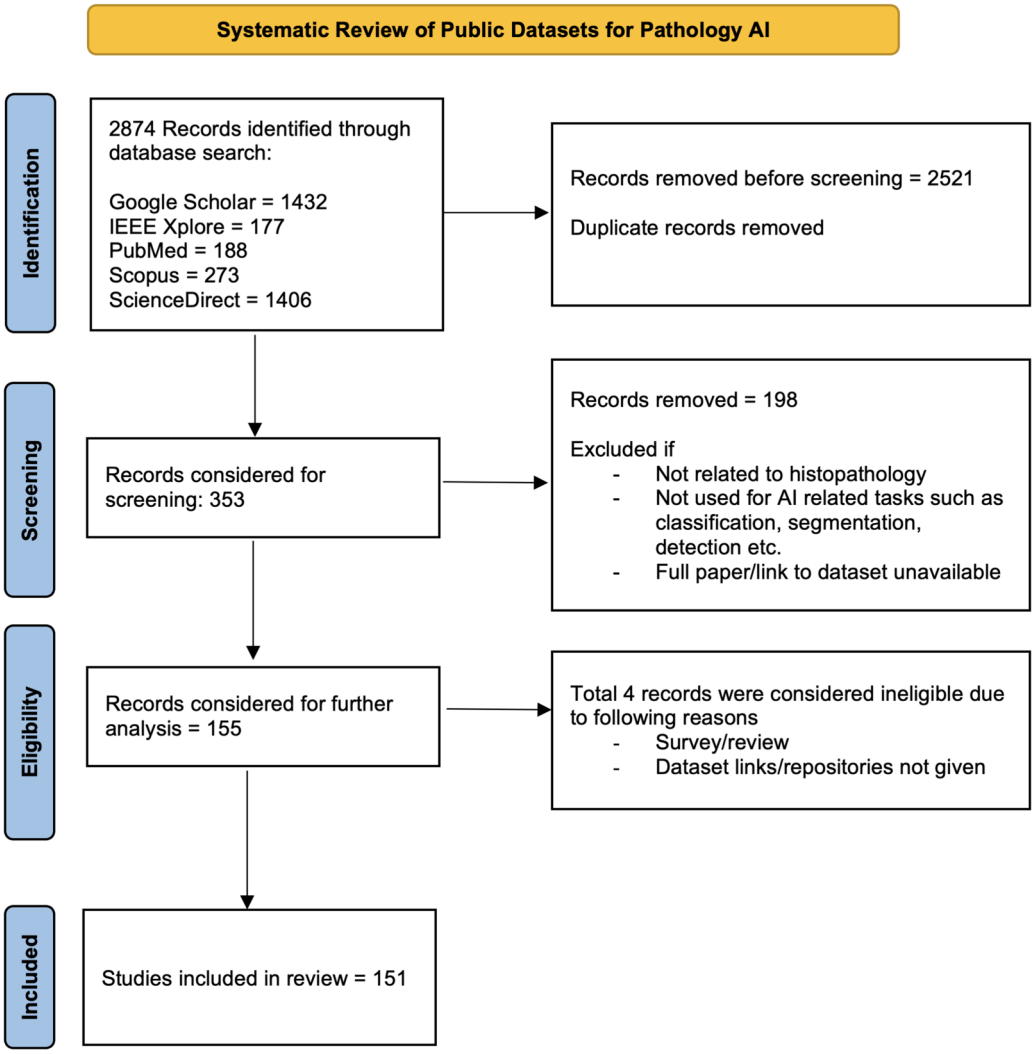

Fig. 1 PRISMA flowchart for our systematic review

pathologists. The images, scanned at 200× magnification, come with detailed clinical data such as tumor type, size, molecular subtype, and metastasis status. Divided into training, validation, and test sets, it facilitates deep learning research, particularly for predicting axillary lymph node (ALN) metastasis. Annotations are stored in JavaScript object notation (JSON) format, and preprocessing code is included for extracting tumor regions.

8. MHIST [28]: MHIST is a dataset of 3152 H&E-stained colorectal polyp images, labeled by expert pathologists. It focuses on the binary classification of hyperplastic polyps (benign) and sessile serrated adenomas (precancerous). The dataset includes image files, annotations, and checksums for data integrity. It supports research into digital pathology, particularly in colorectal polyp classification, and encourages the development of advanced histology image analysis techniques.
9. A histopathological image dataset for grading breast invasive ductal carcinomas [29]: This dataset contains 922 histopathological images of invasive ductal carcinoma (IDC), categorized into three grades based on the modified Bloom Richardson grading system. Images, captured under consistent conditions at multiple magnifications, support AI research in breast cancer grading. Collected from 124 patients, the data is intended for developing machine learning models that analyze tumor features such as nuclear pleomorphism and mitotic count.
10. Histopathological Image based Skin Cancer Classification Using CNN [30]: Collected from 354 patients, this dataset includes 16,099 H&E-stained histology images of benign and malignant skin tumors. The images, captured at different magnifications, are divided into eight benign subclasses and three malignant subclasses,

including melanoma. It supports research in skin cancer diagnosis using CNNs, offering a comprehensive resource for classifying various skin tumor types.

11. Histopathological image patches from colorectal cancer with three classes: tumor, stroma, and others [31]: This dataset consists of 2770 histopathological image patches (224 × 224 pixels) of colorectal cancer tissue. The patches, annotated by a pathologist, are divided into three classes: tumor, stroma, and others (including normal tissue and debris). It can be used as an external test set for deep learning models in colorectal cancer analysis.
12. Histopathological imaging database for Oral Cancer analysis [32]: The dataset contains 1224 images of normal and oral squamous cell carcinoma (OSCC) tissues, captured at 100× and 400× magnifications. Collected from 230 patients, the dataset supports research in oral cancer diagnosis, enabling architectural and cellular-level analysis for developing AI-based computational tools.
13. A histopathological image dataset for endometrial disease diagnosis [33]: This dataset includes 3302 histopathological images of endometrial tissue, categorized into normal endometrium, polyp, hyperplasia, and adenocarcinoma. Collected from patients undergoing hysteroscopic surgery, it facilitates research into endometrial disease diagnosis using deep learning. The dataset provides a robust foundation for training AI models to identify complex morphological features in endometrial tissue.
14. Multi-class texture analysis in colorectal cancer histology [34]: This dataset consists of 5000 histological images derived from 10 anonymized H&E-stained colorectal cancer (CRC) tissue slides. Sourced from the pathology archive at the University Medical Center Mannheim, the dataset includes images of both low- and high-grade tumors. Each image is divided into 150×150-pixel tiles, capturing texture features at different scales, from individual cells to larger mucosal structures. The dataset categorizes images into eight tissue types: tumor epithelium, simple stroma, complex stroma, immune cells, debris, mucosal glands, adipose tissue, and background. Additionally, ten larger images (5000 × 5000 pixels) are included for testing classification methods in more realistic scenarios. It provides a robust resource for studying texture analysis and multi-class classification in CRC histology, aimed at advancing machine learning applications in digital pathology.
15. Invasive Ductal Carcinoma (IDC) Histology Image Dataset [35]: This dataset includes 277,524 image patches (50 × 50 pixels each) extracted from 162 whole-slide images of breast cancer specimens scanned at 40× magnification. The patches are labeled as either IDC-positive (78,786 patches) or IDC-negative (198,738 patches), representing Invasive Ductal Carcinoma and non-IDC tissue. The dataset includes essential metadata such as patient ID and patch coordinates. It was used to train deep learning models to detect IDC, and it allows researchers to replicate the study's training and testing protocols for further analysis and model development. This dataset supports high-resolution patch-level analysis and is valuable for comparing AI models in breast cancer diagnosis.
16. Ovarian Cancer & Subtypes Dataset Histopathology [36]: This dataset contains histopathology images categorized into four subtypes of ovarian cancer—clear cell, endometrioid, mucinous, and serous—as well as non-cancerous images. Organized into five folders, the dataset supports studies on ovarian cancer subtype classification. It is designed for research into both cancerous and non-cancerous ovarian tissue, aiding the development of machine learning models for the diagnosis and classification of ovarian cancer, which is crucial for improving treatment and early detection strategies.
17. ViCE Histopathology Images [37]: The ViCE dataset contains intestinal tissue images designed to facilitate the automated analysis of the villus height-to-crypt depth (Vh:Cd) ratio. This ratio is an essential diagnostic feature for diseases like celiac disease and is often used to evaluate treatment responses. The dataset supports the ViCE software, which automates the quantification of histological parameters, reducing the manual workload in clinical trials. ViCE employs mathematical morphology to analyze tissue geometry and offers a scale-agnostic, user-friendly platform for intestinal disease diagnosis, which can be applied in both research and clinical environments.
18. HEPASS Algorithm Dataset [38]: The HEPASS dataset consists of 385 ,H&E-stained liver histology images from 77 liver donors, aimed at quantifying liver steatosis (fat accumulation), which is crucial for assessing liver graft suitability in transplantation. The dataset is paired with the HEPASS algorithm, a hybrid deep learning framework that combines adaptive thresholding and CNNs to detect and classify micro- and macrosteatosis. Achieving an accuracy of 97.27% on the test set, the algorithm processes images quickly (0.72 s per image) and reliably, surpassing existing steatosis segmentation methods. This dataset advances research in liver pathology and transplantation.
19. RINGS Algorithm Dataset [39]: This dataset contains image data and manual annotations supporting the RINGS algorithm, which is designed for prostate

gland segmentation in histopathological images. Prostate gland segmentation is vital for detecting prostate cancer. RINGS employs a hybrid approach using both traditional image processing and deep learning techniques to improve the accuracy and efficiency of gland segmentation, outperforming seven existing methods with a Dice score of 90.16%. The dataset is invaluable for prostate cancer diagnosis research and aims to reduce the time and errors associated with manual annotations in clinical settings.

20. NDB-UFES Dataset [40]: The NDB-UFES dataset comprises 237 high-resolution histopathological images focusing on oral cancer, specifically oral squamous cell carcinoma (OSCC) and leukoplakia. Images were collected from 2010 to 2021 at the Federal University of Espirito Santo and include patient information such as tobacco and alcohol use. Images were captured at 10× and 40× magnification using optical microscopy. The dataset supports research into AI-driven diagnostic tools for oral cancer, combining image data with socio-demographic factors for a more comprehensive analysis, potentially improving the diagnostic accuracy of clinicians and pathologists.
21. RENFAST Algorithm Dataset [41]: This dataset supports the RENFAST algorithm, an automated method for segmenting kidney blood vessels and fibrosis in histopathological images, crucial for evaluating kidney transplants. RENFAST employs deep learning to segment blood vessels and an adaptive stain separation method for fibrosis detection, achieving high accuracy (0.89 for blood vessels and 0.92 for fibrosis) while processing images in under three seconds. The dataset and algorithm represent significant advances in nephrology, offering quantitative assessments of kidney grafts, which are typically evaluated manually by pathologists.
22. DeepHP Dataset [42]: The DeepHP dataset contains 394,926 image patches from gastric biopsy slides, categorized as Helicobacter pylori positive or negative. The images, stained with H&E and captured at 20× magnification, were used to train CNN models to detect $Helicobacterpylori$ infections. The dataset achieved high accuracy in infection detection, surpassing some endoscopy-based studies, and is valuable for developing deep learning tools for gastric pathology. It provides a large, publicly available dataset for researchers to explore helicobacter pylori detection and other histopathological tasks.
23. HistMNIST [43]: The HistMNIST dataset is a collection of 28 × 28 grayscale image patches from bone marrow biopsies of 16 patients. Originally designed to resemble the MNIST format for machine learning applications, the dataset includes 6800 training images, 2000 validation images, and 2000 test images. It was used to train deep generative models such as variational autoencoders (VAEs). HistMNIST is valuable for researchers working on unsupervised learning in medical imaging, particularly for bone marrow histopathology. The dataset promotes innovations in AI applications for bone marrow biopsy analysis.
24. Histopathology Imagery Dataset of Ph-negative Myeloproliferative Neoplasm [44]: This dataset contains 300 bone marrow trephine images focused on Ph-negative myeloproliferative neoplasms (MPNs), including essential thrombocythemia (ET), polycythemia vera (PV), and primary myelofibrosis (MF). Captured using an Olympus BX41 microscope at magnifications of 10×, 20×, and 40×, the images aim to support AI-based diagnosis in MPNs, enabling applications like image segmentation and disease classification. The dataset addresses diagnostic challenges in rare MPNs and provides a resource for developing tools to assist in early, accurate detection.
25. NCT-CRC-HE-100K [45]: This dataset consists of 100,000 histological image patches (224 × 224 pixels) from colorectal cancer (CRC) and normal tissue, extracted from 86 H&E-stained slides at 0.5 microns per pixel. The dataset includes nine tissue classes such as normal colon mucosa, adipose, smooth muscle, and colorectal adenocarcinoma epithelium, with all images color-normalized using Macenko's method. The dataset provides a large-scale resource for developing automated image analysis algorithms, specifically for colorectal cancer classification.
26. NCT-CRC-HE-100K-NONORM [45]: This version of the NCT-CRC-HE-100K dataset contains the same number of image patches (100,000) and tissue classes as the original dataset, but without color normalization. The absence of normalization preserves natural staining variations, offering a more realistic representation of histological images. This dataset is valuable for studying the impact of color normalization on machine learning models in pathology.
27. CRC-VAL-HE-7K [45]: Designed as a validation set for models trained on the NCT-CRC-HE-100K dataset, CRC-VAL-HE-7K consists of 7180 image patches (224 × 224 pixels) from 50 colorectal adenocarcinoma patients. The set maintains similar specifications to the primary dataset but with no patient overlap, ensuring independence in model testing. It allows researchers to assess the generalizability and accuracy of their colorectal cancer classification algorithms.
28. Histological Image Processing Features Induce a Quantitative Characterization of Chronic Tumor Hypoxia [46]: This dataset includes 66 histological images of colorectal cancer xenografts, stained with H&E and anti-pimonidazole. The images capture different concentrations

of anti-pimonidazole to study hypoxic tumor cells, necrotic tumor cells, and viable tumor cells. This dataset aids in the analysis of tumor hypoxia and the development of methods for quantifying hypoxic features in cancer research.

29. Hematoxylin and eosin stained bladder urothelial cell carcinoma versus inflammation digital histopathology image dataset [47]: The dataset contains 14,891 histopathology images of urinary bladder tissues, focusing on urothelial cell carcinoma (UCC) and inflammation. Captured from 90 H&E-stained slides of 74 patients at 20× magnification using an Olympus E-330 digital camera, the images have a resolution of 3136 × 2352 pixels (JPEG format). It includes 5948 inflammation and 5811 UCC images, with 3132 invalid images excluded. The dataset, manually annotated by a pathologist, is publicly available at Dryad.
30. HistologyHSI-GB [48]: The HistologyHSI-GB dataset contains 469 hyperspectral images (400–1000 nm range, 20× magnification) from glioblastoma biopsies of 13 patients. Each image includes tumor and non-tumor regions, annotated by pathologists to aid in distinguishing these areas. Collected from H&E-stained slides, the dataset provides calibrated HS images, corresponding RGB images, and white/dark references for spectral consistency. This collection supports computational pathology, facilitating glioblastoma analysis through spectral data, image classification, and spectral unmixing, making it valuable for developing diagnostic algorithms in brain tumor research.
31. H&E-stained oral squamous cell carcinoma histological images dataset [49]: This dataset contains 1020 H&E-stained histological images (640 × 640 pixels) of oral squamous cell carcinoma (OSCC), manually annotated by specialists. Captured at 20× magnification, the dataset is suitable for image segmentation tasks and is valuable for advancing research in CAD and deep learning applications focused on oral cancer detection.
32. Gastric Cancer Histopathology Tissue Image Dataset (GCHTID)/HMU-GC-HE-30K [50]: Comprising 31,096 non-overlapping images (224 × 224 pixels) of human gastric cancer tissue, this dataset is derived from H&E-stained slides. It includes eight distinct tissue categories such as cancer-associated stroma, smooth muscle, and gastric tumor tissue. Created to support machine learning research in gastric cancer analysis, GCHTID provides a robust resource for tissue classification and model testing.
33. Enteroscope Biopsy Histopathological H & E Image Dataset (EBHI-Seg) [51]: This dataset contains 4456 H&E-stained colorectal biopsy images, each with corresponding ground truth annotations for image segmentation tasks. It covers six tumor differentiation stages, including normal tissue, polyps, and adenocarcinoma, offering a resource for developing advanced segmentation algorithms in colorectal cancer diagnosis.
34. Automatic Registration Of Breast Cancer Tissue [52]: The ACROBAT 2023 dataset comprises WSIs of breast cancer tissue sections with matched H&E and immunohistochemistry (IHC) stains. The dataset includes 750 development cases, 100 validation cases, and 200 test cases, providing a resource for developing image registration algorithms across different stains. It focuses on evaluating algorithm performance under domain shifts and supports research in breast cancer pathology.
35. Automatic Non-rigid Histological Image Registration (ANHIR) Dataset [53]: ANHIR is a collection of high-resolution whole-slide images designed for non-linear image registration in digital pathology. It includes tissue samples stained with different dyes and accompanied by manually annotated landmarks to evaluate registration quality. This dataset challenges researchers to develop robust, automatic registration methods for histopathological images with complex textures and varied stains.
36. ICIAR 2018 BACH Dataset [54]: The ICIAR 2018 BACH dataset includes 400 microscopy images and whole-slide images of breast histology, annotated into four classes: normal, benign, in situ carcinoma, and invasive carcinoma. The dataset supports classification and segmentation tasks in breast cancer diagnosis, providing a valuable resource for developing AI-based diagnostic tools.
37. BRACS [55]: The BRACS dataset contains 547 WSIs and 4539 regions of interest (ROIs) from breast cancer histopathology images, categorized into seven lesion subtypes such as benign lesions, usual ductal hyperplasia, and ductal carcinoma in situ. It includes atypical lesions, which are often overlooked in other datasets, making it a valuable resource for developing and evaluating AI algorithms for breast cancer diagnosis.
38. CAMELYON16 [56]: The CAMELYON16 dataset is a pioneering collection of H&E-stained WSIs aimed at detecting metastases in breast cancer patients' sentinel lymph nodes. The dataset, sourced from two Dutch medical centers, includes both training and test sets for algorithm development in digital pathology. As the first challenge of its kind, CAMELYON16 spurred innovation in using WSIs for metastasis detection, attracting global research interest. This dataset played a crucial role in advancing AI-driven diagnostic tools and reducing subjectivity in cancer diagnosis, especially for sentinel lymph node evaluations.
39. CAMELYON17 [57]: CAMELYON17 builds on the CAMELYON16 dataset, comprising 899 WSIs of lymph node sections from breast cancer patients collected from five medical centers. It is labeled according to the TNM

classification system, focusing on three metastasis categories: isolated tumor cells (ITC), micrometastases, and macrometastases. The dataset includes detailed annotations in extensible markup language (XML) and comma-separated values (CSV) formats, making it an invaluable resource for developing AI models aimed at improving breast cancer diagnosis and lymph node metastasis detection.

40. CPTAC-BRCA [58]: The CPTAC-BRCA dataset centers on breast invasive carcinoma and contains histopathology data for 134 patients, along with clinical, genomic, and proteomic information. With a total data size of 113.29 GB, this dataset integrates multi-omic and clinical data, allowing researchers to study cancer phenotypes and their correlations with molecular profiles, aiding cancer research through comprehensive data.
41. DRYAD dataset (HASHI) [59]: The HASHI dataset, available via Dryad, supports the High-throughput Adaptive Sampling for Histopathology Image analysis (HASHI) method, designed for processing large WSIs of breast cancer with CNNs. Covering 500 training and 195 test cases, the dataset includes XML annotations and binary masks for robust segmentation and classification. HASHI addresses the computational demands of WSIs, helping reduce sample size while maintaining accuracy, particularly for invasive breast cancer detection.
42. GTEx Portal [60]: The GTEx database is a publicly accessible repository that integrates genomic, transcriptomic, and clinical data from over 900 donors across approximately 50 tissue types, including hard-to-access organs like the brain, heart, and pancreas. It enables researchers to explore genetic variation and gene expression relationships through genotype data, RNA sequencing, and expression quantitative trait loci (eQTL) mappings. Hosted on platforms like NCBI and dbGaP, it offers both open and controlled access to protect donor privacy. GTEx supports analyses of cis- and trans-eQTLs, allele-specific expression, alternative splicing, and gene regulatory networks. Linked with resources like ENCODE, it facilitates multi-layered studies in epigenomics, proteomics, and disease mechanisms. Researchers can access visualization tools, statistical pipelines, and protocols through the GTEx Portal, making it a vital resource for functional genomics and precision medicine.
43. HER2-Warwick dataset [61]: Created for the HER2 Scoring Contest in 2016, this dataset consists of approximately 100 WSIs used to automate HER2 scoring in breast cancer. Participants developed algorithms to assess HER2 status from H&E-stained slides, aiming to reduce subjectivity in breast cancer diagnosis. With the large image dimensions (100,000 × 80,000 pixels) and ground truth HER2 scores provided by pathologists, the dataset offers a unique challenge for developing accurate, automated diagnostic tools.
44. HEROHE [62]: The HEROHE Challenge dataset consists of 509 WSIs of breast cancer tissue, stained with H&E. These slides are linked to HER2 classifications derived from IHC and in situ hybridization (ISH) testing. Researchers use this dataset to develop algorithms predicting HER2 status based solely on morphological features, making it a valuable tool for advancing breast cancer diagnostics in computational pathology.
45. IMPRESS dataset: This dataset includes 126 WSIs of breast cancer tissue from patients with HER2-positive and triple-negative breast cancer, captured after neoadjuvant chemotherapy (NAT). It also includes multiplex IHC-stained slides featuring biomarkers such as PD-L1 and CD8, as well as corresponding clinical data. This dataset supports AI-based analysis for predicting chemotherapy outcomes, aiding in personalized treatment strategies.
46. Post-NAT-BRCA [63]: Comprising 96 high-resolution WSIs from breast cancer patients who underwent NAT, this dataset focuses on residual invasive breast cancer. It includes annotations for tumor cellularity, necessary for calculating the residual cancer burden index (RCBi), along with clinical data to support cell segmentation and treatment outcome prediction algorithms.
47. SLN-Breast [64]: This dataset contains 130 WSIs of axillary lymph node specimens from 78 breast cancer patients, scanned at 20× magnification. Of these, 36 slides show metastatic breast carcinoma. The dataset includes binary class labels indicating the presence or absence of metastases, making it suitable for developing models to detect lymph node metastases in breast cancer patients.
48. TCGA-BRCA Dataset [65]: The TCGA-BRCA dataset is part of the Cancer Genome Atlas (TCGA) and includes clinical, genetic, and pathological data for 139 breast cancer patients, along with 3111 H&E-stained WSIs. It is the largest publicly available breast cancer histopathological dataset and is designed for research into the relationship between breast cancer phenotypes and genotypes, supporting AI-driven diagnostic and prognostic tools.
49. TIGER Dataset [66]: This dataset focuses on quantifying Tumor-Infiltrating Lymphocytes (TILs) in breast cancer histopathology, particularly for HER2+ and Triple-Negative Breast Cancer (TNBC). It includes 195 WSIs from TCGA-BRCA and other medical institutions, annotated for seven tissue regions and lymphocyte/plasma cell bounding boxes. This dataset is ideal for developing AI-based prognostic models for TIL assessment in cancer research.

50. GasHisSDB [67]: The GasHisSDB dataset includes 245,196 sub-size images of gastric histopathology, designed for developing computer-aided diagnosis algorithms. The images are divided into normal and abnormal classes, with multiple resolutions for weakly supervised learning. Labeled by experienced pathologists, this dataset provides a large-scale resource for machine learning research in gastric cancer detection.
51. MIHIC [68]: The MIHIC dataset contains 309,698 image patches from lung cancer slides stained with 12 IHC markers (e.g., CD3, CD20). Manually annotated by pathologists, the dataset focuses on tumor immune microenvironment quantification, making it a valuable resource for developing deep learning models for lung cancer prognosis and immune microenvironment research.
52. BMIRDS Dartmouth Lung Cancer Histology Dataset [69]: This dataset contains 143 WSIs of lung adenocarcinoma, sourced from Dartmouth-Hitchcock Medical Center. Labeled by three pathologists for predominant histological patterns (e.g., Lepidic, Acinar), it is designed to facilitate research in lung cancer classification and histopathological analysis, supporting the development of AI models for lung cancer detection and diagnosis.
53. BMIRDS Dartmouth Kidney Cancer Histology Dataset [70]: This dataset includes 563 de-identified WSIs of renal cell carcinoma (RCC) sourced from the Department of Pathology at Dartmouth-Hitchcock Medical Center. These H&E-stained formalin-fixed paraffin-embedded (FFPE) slides are labeled for various RCC subtypes, including Renal Oncocytoma, Chromophobe RCC, Clear Cell RCC, and Papillary RCC. The dataset is divided into training, validation, and testing sets, supporting research in RCC classification and histological image analysis.
54. Gland Segmentation in Colon Histology Images (GlaS) [71]: The GlaS dataset consists of 165 H&E-stained histological images of colorectal adenocarcinoma, annotated for gland boundaries. The dataset includes both benign and malignant tissue samples, facilitating the development of automated gland segmentation algorithms for cancer grading. This dataset was used in the GlaS Challenge at MICCAI 2015, helping researchers advance colorectal cancer image analysis.
55. ACDC-LungHP [72]: The ACDC-LungHP dataset is part of a challenge designed for detecting and classifying lung carcinoma in WSIs. It includes 200 annotated biopsy samples of H&E-stained lung tissue and focuses on segmenting lung cancer regions and classifying subtypes. The dataset supports research into AI systems for detecting lung cancer, a leading cause of cancer-related deaths.
56. Count-ception [73]: This dataset, part of the Count-ception project, focuses on cell counting in various types of cell imagery, including adipocytes, bone marrow cells (MBM), and bacterial cells (VGG). VGG Cells consist of 200 images with simulated bacterial cells generated through fluorescence-light microscopy, containing about 174 cells per image. MBM Cells are based on real bone marrow samples from healthy individuals, consisting of 44 images with an average of 126 cells, where the dataset was refined to improve accuracy. The Adipocyte Cells dataset is derived from histology slides of human subcutaneous adipose tissue, containing 200 regions of interest with about 165 cells per image.
57. ADP Dataset [74]: The ADP dataset comprises WSIs from various organs, including brain, kidney, breast, liver, and heart. Digitized at 40× magnification, these 1088 × 1088-pixel image patches exclude background regions and are labeled by histological type. This diverse dataset is designed to facilitate the development of computational pathology algorithms for tissue classification and disease diagnosis across multiple organs.
58. Automated Gleason Grading Challenge 2022 (AGGC) [75]: This dataset includes H&E-stained WSIs of prostate specimens, annotated for Gleason patterns by expert pathologists. It is sourced from the National University Hospital in Singapore and scanned using various scanners. The dataset includes 105 prostatectomy and 37 biopsy cases for training, and 45 prostatectomy and 16 biopsy cases for testing, supporting research in automated Gleason grading for prostate cancer diagnosis.
59. Multi-Class Cell Detection Using Spatial Context Representation (BRCA-M2C) [76]: This dataset supports multi-class cell detection and classification in breast cancer, lung cancer, and colorectal cancer. It includes 120 patches from TCGA and surveillance, epidemiology, and end results (SEER) cohorts, with 500 × 500 pixel image patches annotated for inflammatory, epithelial, and stromal cells. The dataset is valuable for joint cell detection and classification research in cancer histopathology.
60. HER2 Tumor ROIs [77]: This dataset includes 273 WSIs of HER2+ breast cancer with manually annotated tumor ROIs. It contains both HER2+ and HER2− samples from Yale and TCGA, aiding in the development of computational models for predicting HER2 status and response to trastuzumab treatment using H&E-stained slides.
61. CryoNuSeg [78]: CryoNuSeg is a fully annotated dataset for nuclei segmentation in cryosectioned H&E-stained histological images. The dataset includes 30 images from ten human organs, with manual segmentations by two experts. It is designed to support the development of deep learning models for nuclei seg-

mentation, particularly addressing challenges posed by cryosection versus FFPE processing.

62. Lizard [79]: The Lizard dataset focuses on nuclear instance segmentation and classification in colonic tissue. It includes nearly 500,000 annotated nuclei from normal, inflammatory, dysplastic, and cancerous states. The dataset is designed to support the development of computational models for tumor microenvironment analysis and colonic tissue profiling.
63. Pan-tumor T-lymphocyte Dataset [80]: This dataset includes IHC-stained slides for CD3+ T-lymphocyte detection in head and neck, lung, breast, and gastric cancers. The images are annotated with bounding boxes for T-lymphocytes and other cell types, supporting research in tumor immune microenvironment analysis.
64. CoNSeP - HoVer-Net [81]: The CoNSeP dataset consists of 41 H&E-stained image tiles from colorectal adenocarcinoma, annotated for nuclear segmentation and classification. The dataset includes over 24,000 nuclei annotations and supports simultaneous nuclear segmentation and classification, facilitating computational pathology research.
65. PANDA Challenge Dataset [82]: The PANDA challenge dataset comprises 12,625 WSIs of prostate biopsies from Europe and the USA, supporting research in prostate cancer diagnosis and Gleason grading. The dataset is diverse, spanning multiple populations, laboratories, and scanning equipment, allowing for the development of AI models with generalizability across various cohorts.
66. PanNuke [83]: The PanNuke dataset includes almost 200,000 segmented nuclei across five key cell classes (neoplastic, non-neoplastic epithelial, connective, inflammatory, and dead cells) from 19 different tissue types. The annotations, created using a semi-automated pipeline and reviewed by pathologists, ensure accuracy and clinical relevance. The dataset contains over 2000 visual fields and addresses challenges like artifacts and variations in WSIs. PanNuke is valuable for training models to segment and classify nuclei in diverse tissue contexts, improving model generalization to real-world clinical scenarios.
67. Janowczyk et al [35]: This dataset consists of 143 estrogen receptor-positive (ER+) breast cancer images scanned at 40× magnification, each image being 2000 × 2000 pixels. Approximately 12,000 nuclei were manually segmented, and both the original H&E-stained images and their corresponding nuclei masks (with white pixels representing nuclei regions) are provided. The dataset is ideal for training deep learning models for nuclei segmentation and captures complex variations in nuclei texture, morphology, and size, making it suitable for robust segmentation model testing.
68. NuCLS [84]: NuCLS is a large-scale dataset for nucleus classification, localization, and segmentation, primarily focused on breast cancer histopathology. It contains over 220,000 annotations from crowdsourcing and expert pathologists across 3944 fields of view (FOVs) from 125 triple-negative breast cancer patients. It provides both single-rater and multi-rater annotations, offering valuable ground truth data for training and benchmarking models in nucleus detection, segmentation, and classification. It bridges the gap between limited expert annotations and scalable data generation for computational pathology.
69. Bone-Marrow-Cytomorphology [85]: This dataset includes over 170,000 de-identified bone marrow cell images from 945 patients, focused on diagnosing hematologic malignancies like leukemia and lymphoma. The images were captured using brightfield microscopy with May-Grünwald-Giemsa/Pappenheim stain at 40× magnification with oil immersion. It is one of the largest publicly available bone marrow cytology collections and serves as a crucial resource for developing AI-driven diagnostic algorithms to detect hematologic diseases.
70. PAIP2021: The PAIP2021 dataset focuses on detecting perineural invasion (PNI) in cancerous tissues across multiple organs. It includes high-resolution WSIs at 20× and 5× magnifications with detailed annotations, labeled as PNI (nerve, tumor, benign) or non-PNI. This dataset aids in developing deep learning models for automated PNI detection, an important factor in cancer pathology, and helps improve treatment decision-making by providing insights into nerve involvement in tumors.
71. WSSS4LUAD [86]: This dataset is part of the weakly supervised tissue semantic segmentation challenge for lung adenocarcinoma. It comprises 67 WSIs (49 from Guangdong Provincial People's Hospital and 14 from TCGA) and contains image-level annotations for training and pixel-level annotations for validation and testing. It focuses on tumor epithelial tissue, tumor-associated stroma, and normal tissue, and supports research into weakly supervised methods for segmentation.
72. CellSeg [87]: This dataset is used to train the CellSeg software, a Mask R-CNN-based tool for nucleus segmentation. It contains 29,464 manually segmented nuclei from various tissue microenvironments, particularly from cancer research. The dataset includes biological microscopy images with diverse cell phenotypes, sizes, and brightness conditions, ensuring robustness across different imaging conditions and supporting deep learning research for nucleus segmentation.

73. PAIP2023: The PAIP2023 contains 100 high-resolution WSIs of liver tissue from South Korean medical centers, focused on Hepatocellular Carcinoma (HCC) tumor segmentation. Scanned at 20× magnification, the dataset includes pixel-level ground truth masks for the training and validation sets. This dataset supports advanced segmentation research in HCC, addressing the complexity of tumor boundaries in large pyramidal images.
74. NuClick [88]: NuClick consists of datasets for nuclear, cell, and gland segmentation, including the GlaS and CRAG datasets for gland segmentation, MonuSeg and CPM for nuclear segmentation, and synthetic datasets for white blood cell (WBC) counting. This dataset facilitates robust model training and evaluation for segmentation tasks across different biological structures.
75. MIDOG 2022 [89]: MIDOG 2022 is a dataset designed for mitotic figure detection in a wide range of tumor types across different species (human, canine, feline). It includes 405 tumor cases and 9501 annotated mitotic figures from multiple domains, captured using various whole-slide image scanners. It supports the development of robust deep learning models for mitotic figure detection, which are critical for assessing tumor aggressiveness.
76. MIDOG 2021 [90]: This dataset focuses on mitotic figure detection in breast cancer histopathology images, emphasizing domain generalization across different imaging scanners. It includes WSIs from 50 breast cancer cases per scanner, with annotations for mitotic figures and non-mitotic examples. The dataset supports research on detecting mitotic figures and adapting algorithms to various scanning devices.
77. MIDOG++ [91]: MIDOG++ is a multi-domain dataset for mitotic figure detection in human and canine tumors. It includes 503 histological specimens and annotations for 11,937 mitotic figures, representing variability in tumor types, laboratories, and whole-slide image scanners. The dataset supports the development of deep learning algorithms for robust mitotic figure detection across different tumor types and domains.
78. CAMEL [92]: The CAMEL dataset, derived from the CAMELYON16 dataset, focuses on histopathology image segmentation using weak supervision. It contains 400 WSIs of sentinel lymph nodes stained with H&E, split into patches for training and testing. CAMEL supports weakly supervised learning by providing image-level labels and refining them to instance-level segmentation for cancer detection.
79. OCELOT [93]: The OCELOT dataset contains small and large field-of-view patches from TCGA WSIs, stained with H&E. It includes annotations for both cells and tissues, supporting cell detection and tissue segmentation research. The dataset provides detailed spatial information, with rigorous multi-pathologist annotations, aiding research in computational pathology.
80. CAMELYON [94]: CAMELYON comprises 1399 WSIs of sentinel lymph node sections from breast cancer patients, aimed at advancing machine learning models for detecting cancer metastases. It includes detailed metastasis annotations and was used in the CAMELYON16 and CAMELYON17 challenges, promoting research in CNNs for metastasis detection.
81. BCSS [95]: This dataset contains 151 WSIs of TNBC from TCGA, with over 20,000 annotated regions (tumor, stroma, lymphocytes). It is designed to support the development of deep learning models for semantic segmentation in cancer pathology, particularly for training models that improve breast cancer diagnosis and treatment.
82. SegPath [96]: SegPath offers a large-scale dataset for histology segmentation, with annotations for eight major cell types across multiple organs and cancers. It provides high-throughput and accurate tissue/cell-type segmentation, improving on manual annotations by including a wider range of cell morphologies. It is designed to advance cancer diagnosis and research through machine learning.
83. SPIE-AAPM_NCI BreastPathQ [97]: The BreastPathQ dataset includes 96 WSIs of breast cancer histology from 55 patients treated with neoadjuvant therapy. It contains 2394 image patches (512 × 512 pixels) annotated for tumor cellularity by expert pathologists. The dataset supports developing models for assessing tumor cellularity, an important factor in determining treatment response.
84. NADT-Prostate [98]: This dataset includes histopathology images and clinical data from 37 prostate cancer patients undergoing neoadjuvant therapy. It contains tissue slide images with various IHC stains, as well as genomic and clinical information, making it a valuable resource for studying treatment response mechanisms in prostate cancer.
85. TIL-WSI-TCGA [99]: The TIL-WSI-TCGA dataset provides TIL maps for over 5200 diagnostic H&E slides from 13 cancer types. Generated using deep learning, the dataset allows for detailed analysis of lymphocyte infiltration patterns, supporting research into the tumor immune microenvironment and its relationship with cancer outcomes.
86. TUPAC16 [100]: A dataset with 821 WSIs to evaluate tumor proliferation. It includes a training set (500 images) and testing set (321 images), along with mitotic scores and PAM50 gene expression-based proliferation scores for each case. Auxiliary ROI and

mitosis detection datasets enhance model training for mitotic figure detection. Variability in tissue staining and domain differences across labs add complexity, making this dataset a valuable real-world resource for developing models that assess tumor aggressiveness through proliferation metrics.

87. TUPAC16 – aux [100]: The Tumor Proliferation Assessment Challenge (TUPAC16) provides 500 breast cancer cases from TCGA, each annotated with proliferation scores based on mitosis counting and molecular data. WSIs are offered in high-resolution .svs format, with auxiliary datasets available for mitosis detection (73 cases) and regions of interest (148 cases). These auxiliary datasets can enhance algorithm development for tumor proliferation prediction, particularly in cases where proliferation correlates with tissue structure and cell nuclei appearance.
88. UniToPatho [101]: This dataset focuses on colorectal polyps and adenoma dysplasia. It includes 9536 H&E-stained image patches derived from 292 whole-slide images (WSIs). The dataset supports training deep neural networks to classify colorectal polyps and assess dysplasia grades, addressing variability in pathologist diagnoses. It aids in developing reliable automated systems for early carcinoma detection and improved colorectal cancer management.
89. Artificial Intelligence for Tumor Tissue Detection and Histological Regression Grading in Esophageal Adenocarcinomas [102]: This dataset contains histological slides from patients with esophageal adenocarcinoma, sourced from pathology institutes in Germany, Austria, and TCGA. The dataset is annotated into 11 tissue classes and includes neoadjuvant-treated and treatment-naive cases. It supports the development of AI algorithms to detect tumor tissues and assess tumor regression, enhancing diagnostic accuracy in esophageal cancer.
90. VisioMel: VisioMel contains WSIs of skin melanomas from French medical centers, provided in a pyramidal TIF format. Each slide is linked to clinical metadata (e.g., patient age, sex, tumor characteristics), including data on melanoma relapse. The dataset is designed to develop predictive models for assessing the likelihood of melanoma relapse, contributing to advances in digital pathology and AI-driven diagnostics in melanoma prognosis.
91. BreCaHAD [103]: BreCaHAD consists of 162 breast cancer histopathology images, annotated into six categories: mitosis, apoptosis, tumor nuclei, non-tumor nuclei, tubule, and non-tubule. The images reflect real-world clinical challenges, such as artifacts and staining variability. The dataset is used to develop and test automated diagnostic tools for breast cancer detection and classification, providing a foundation for machine learning research in breast cancer histopathology.
92. UCSB Bio-Segmentation Benchmark dataset [104]: The UCSB Bio-Segmentation dataset includes a wide range of biological images for evaluating bio-image analysis algorithms. The dataset covers subcellular, cellular, and tissue-level structures in 2D, 3D, and time-lapse sequences. It features images of microtubules, nuclei, breast cancer cells, kidney cells, and retinal tissues from multiple species and imaging modalities. Ground truth annotations are provided for segmentation, classification, and tracking, supporting algorithm benchmarking in biological image analysis.
93. PathVQA database [105]: PathVQA consists of 4998 pathology images and 32,799 associated question-answer pairs. The dataset is designed for visual question answering (VQA) research in pathology, with questions covering "what," "where," "when," "how," and "yes/no" types. Questions were generated using natural language processing techniques from image captions, offering a unique challenge for AI models to reason about pathology images.
94. CoNIC 2022 [106]: CoNIC contains 4981 histology image patches from colon tissue, with half a million labeled nuclei across six types: epithelial cells, connective tissue cells, lymphocytes, plasma cells, neutrophils, and eosinophils. Each patch includes instance segmentation and classification maps. The dataset drives research in nuclear segmentation, classification, and quantification, with applications in colon tissue analysis and pathology under normal, inflammatory, and cancerous conditions.
95. MoNuSAC 2020 [107]: The MoNuSAC 2020 dataset is designed for nuclei segmentation and classification across four organs—lung, prostate, kidney, and breast. These H&E-stained images, digitized at 40× magnification, are sourced from various hospitals and are available in .svs and .tif formats. The training set includes 31,000 nuclear boundary annotations in XML format, while the test set contains images and binary masks of ambiguous regions. The dataset, which supports segmentation and classification tasks in computational pathology, includes expert pathologist annotations and is released under the CC BY-NC-SA 4.0 license.
96. MoNuSeg [108]: MoNuSeg is a nuclei segmentation dataset that provides H&E-stained tissue images from multiple patients with tumors in various organs. Collected from different hospitals, the dataset reflects the diversity of nuclei appearances due to the variety of organs and staining protocols. The training data consists of 30 images with 22,000 nuclear boundary annotations, while the test set adds 7000 annotations. Released under the CC BY-NC-SA 4.0 license, this dataset aims

to facilitate the development of generalizable nuclei segmentation techniques in computational pathology.

97. ARCH [109]: The ARCH dataset is designed for multiple instance captioning in computational pathology, containing over 11,800 image bags and 15,164 individual images paired with detailed captions extracted from pathology textbooks and clinical articles. These captions describe diagnostic and morphological details, covering various tissue types, stains, and pathologies. The dataset supports the learning of transferable representations in pathology, potentially outperforming traditional pre-trained models like ImageNet.
98. Osteosarcoma-Tumor-Assessment [110]: This dataset comprises 1144 H&E-stained histology images of osteosarcoma, collected from four patients at Children's Medical Center in Dallas. The images are annotated into three categories: non-tumor, necrotic tumor, and viable tumor. Each image, sized at 1024 × 1024 pixels, is available at 10× magnification. The dataset helps in the assessment of osteosarcoma, primarily affecting adolescents, and is intended for use in computer-aided diagnosis and tumor analysis.
99. CoCaHis [111]: The CoCaHis dataset includes 82 H&E-stained frozen sections from 19 patients diagnosed with metastatic colon cancer in the liver. The dataset is annotated at the pixel level by a team of four pathologists and includes stain-normalized versions to address batch effects. The dataset has been used to train both traditional machine learning models (e.g., support vector machine (SVM), K-nearest neighbor (KNN)) and deep learning models (e.g., U-Net, DeepLabv3), with deep learning approaches outperforming traditional ones in terms of accuracy and F1 score. It is intended for developing CAD tools for intraoperative pathology analysis.
100. Naylor et al. [112]: This dataset includes two main sources: 50 H&E-stained images of TNBC from the Curie Institute, with 4022 annotated nuclei, and 30 images from the Indian Institute of Technology Guwahati, with 21,623 annotated nuclei from different cancer types. Both datasets were captured at 40× magnification. They feature diverse annotations, including normal epithelial cells, invasive carcinomatous cells, fibroblasts, and immune cells, supporting the development of robust nuclei segmentation models.
101. SICAPv2 [113]: The SICAPv2 dataset focuses on prostate cancer, comprising 155 biopsies from 95 patients. Expert urogenital pathologists annotated the dataset with individual Gleason grades and cribriform patterns, critical for cancer diagnosis and prognosis. The WSIs were digitized at 40× magnification, then downsampled to 10× resolution. Patches of 512 × 512 pixels were extracted and labeled according to the predominant Gleason grade. The dataset supports deep learning research in prostate cancer grading and pattern detection.
102. CPTAC-COAD [114]: Part of the Clinical Proteomic Tumor Analysis Consortium, CPTAC-COAD includes histopathology images from 106 subjects diagnosed with colon adenocarcinoma. This dataset, available on the Cancer Imaging Archive (TCIA), links cancer phenotypes with molecular data and supports the investigation of cancer progression and molecular correlations in colon adenocarcinoma.
103. PAIP2020 [115]: The PAIP2020 dataset focuses on colon adenocarcinoma diagnosis, providing WSIs from patients collected from Seoul National University Hospital. Scanned at 40× magnification, the dataset is split into training, validation, and test sets, with detailed tumor annotations and microsatellite instability (MSI) classification. It supports research into automated colorectal cancer detection and classification using AI.
104. KIMIA Path24C [116]: The Kimia Path24C dataset is designed for classification and retrieval tasks in digital pathology. It includes RGB patches extracted from 24 WSIs of various stains like H&E, IHC, and Masson's trichrome. The patches exclude irrelevant regions like background pixels, ensuring high-quality training data for AI models. The dataset supports research in content-based image retrieval (CBIR) and CAD in pathology.
105. UPENN-GBM [117]: The UPENN-GBM dataset comprises multi-parametric MRI scans and histopathology images from 630 glioblastoma multiforme (GBM) patients treated at the University of Pennsylvania, USA. The dataset includes MRI images in DICOM and NIFTI formats, tumor segmentation labels, and H&E-stained tissue sections. It enables research in tumor progression, radiogenomics, and response to treatment in GBM.
106. Prostate Fused-MRI-Pathology [118]: This dataset includes MRI scans and digitized histopathology images from 28 prostate cancer patients. Scanned using a 3 Tesla MRI, the images include multiple sequences (T1, T2, DWI, DCE) and whole-mount prostatectomy specimens. The dataset facilitates spatial correlation analysis between MRI and pathology data, supporting the development of advanced image analysis algorithms for prostate cancer detection and characterization.
107. PAIP2019 [119]: The PAIP2019 dataset focuses on liver cancer segmentation, particularly HCC. It provides WSIs from liver tissue scanned at 20× magnification, with detailed annotations of tumor boundaries. The dataset is intended for training AI models to accurately detect and quantify tumor regions in liver cancer histopathology.
108. LYON19 [120]: The LYON19 dataset is part of the LYON19 Grand Challenge, aimed at advancing the analysis of lymphocyte distribution patterns in breast,

colon, and prostate tissues using IHC-stained samples. It contains 441 ROIs extracted from WSIs sourced from eight medical centers in the Netherlands. The tissue samples were stained with antibodies against CD3 or CD8, commonly used to highlight lymphocytes. The images were digitized with the Pannoramic 250Flash II scanner at 0.24 μm/px resolution. The dataset presents a wide range of lymphocyte patterns, including regular, clustered distributions, and tissue artifacts, making it ideal for studying variations in lymphocyte behavior.

109. DigestPath2019 [121, 122]: The dataset is designed to aid in detecting signet ring cell carcinoma, a rare and aggressive adenocarcinoma, and segmenting colonoscopy tissues. The dataset includes 77 positive images (containing signet ring cells) and 383 negative images, extracted from whole-slide images of gastric mucosa and intestinal tissues. The images are large (2000 × 2000 pixels) and annotated with tight bounding boxes by pathologists. However, some cells may be missed due to the challenges of manual annotation. An evaluation set with 227 images is also provided to help participants develop robust detection algorithms—with 27 positive and 200 negative images. The Colonoscopy Tissue Segment Dataset is used to automate the segmentation of colonoscopy tissue slices, focusing on differentiating benign and malignant lesions. This dataset includes 250 positive images with pixel-level annotations for malignant regions and 410 negative images without any lesions. An evaluation set with 212 tissues is provided—with 90 positive (containing lesions) and 122 negative tissues. The dataset poses challenges due to variations in image appearance, collected from different medical centers.
110. The Digital Brain Tumour Atlas, an open histopathology resource [123]: The Digital Brain Tumour Atlas is a comprehensive collection of 3115 digitized H&E-stained slides from 2880 patients, spanning 126 brain tumor types from 1995 to 2019. Digitized at high resolution using a Hamamatsu NanoZoomer 2.0 HT slide scanner, the slides underwent expert neuropathologist review to ensure diagnostic quality. The dataset includes clinical annotations such as patient demographics, tumor location, recurrence, and World Health Organization (WHO) classification, with rare tumor types also represented for broader research applications. MATLAB scripts for tissue area and cellularity estimation support advanced computational analysis. The data, available via EBRAINS under ethical non-commercial guidelines, is invaluable for machine learning, diagnostic research, and teaching.
111. Multi-Scanner SCC [124]: This dataset includes histopathological images of canine cutaneous squamous cell carcinoma (SCC), captured using five different slide scanning systems (NanoZoomer, Aperio, and Pannoramic). The dataset contains 220 WSIs with 1243 polygon annotations, covering seven histologic classes like tumor, epidermis, dermis, and inflammation/necrosis. This dataset highlights scanner-induced domain shifts and provides valuable resources for developing models that account for variability in image quality caused by different scanners. This diversity allows for the testing of model robustness in digital pathology segmentation tasks.
112. Gleason_CNN [125]: This dataset comprises tissue microarrays (TMAs) from prostate cancer patients, digitized at 40× magnification. Five TMAs, with around 200–300 spots each, are annotated by expert pathologists for Gleason grading (patterns 3, 4, 5), and benign regions are also marked. The dataset supports Gleason pattern classification and prostate cancer diagnosis using machine learning models. The data also includes clinical information and survival outcomes for three TMAs, providing a comprehensive foundation for prostate cancer research.
113. MITOS_WSI_CMCv [126]: The dataset contains 21 WSIs of canine mammary carcinoma (CMC), with 13,907 mitotic figures and 36,379 mitotic-like figures annotated by expert pathologists. The dataset, scanned at 0.25µm/px, serves as a valuable resource for mitotic activity detection and classification in cancer grading, applicable in both veterinary and human breast cancer research. The detailed annotations enable the development of machine learning models to improve cancer diagnosis and grading.
114. IMP-CRS 2024 [127]: This dataset consists of 5333 WSIs of colorectal biopsy and polypectomy samples. The slides, digitized at 40× resolution, are categorized into non-neoplastic, low-grade, and high-grade lesions, including invasive adenocarcinomas. The dataset, totaling about 5 terabytes (Tb) of data in .svs format, is intended for training machine learning models in digital pathology, particularly focusing on colorectal cancer diagnosis. The size and variety of the dataset provide a robust platform for developing AI-based diagnostic tools.
115. CRCHisto [128]: This dataset contains 100 histology images of colorectal adenocarcinomas, each sized 500 × 500 pixels with a resolution of 0.55μm/px. The dataset includes annotations for over 29,000 nuclei, classified into epithelial, inflammatory, fibroblast, and miscellaneous categories. Expert pathologists conducted the annotations, ensuring a high level of accuracy for nuclei segmentation and classification. This dataset is particularly useful for developing models in colorectal cancer pathology.

116. CPTAC-OV [129]: The CPTAC-OV dataset is part of the National Cancer Institute's CPTAC and focuses on ovarian serous cystadenocarcinoma. The dataset includes histopathology images from 102 human subjects, amounting to 54.64 GB of data in SVS format. These images, available through TCIA, are intended to support cancer phenotyping research by correlating imaging data with proteomic, genomic, and clinical data, facilitating advancements in ovarian cancer research.
117. ENDO-AID [130]: This dataset consists of 91 WSIs of endometrial carcinoma samples obtained via minimally invasive Pipelle biopsies. The slides, stained with H&E and scanned at 0.25 μm/px, are available in TIFF format at 0.5 μm/px, facilitating compatibility with various pathology viewers. Developed by Radboud University Medical Center, Netherlands. ENDO-AID supports benchmarking on a grand-challenge.org platform, allowing researchers to evaluate model performance against pathologist assessments in endometrial carcinoma diagnosis.
118. Digital Pathology Dataset for Breast Cancer Diagnosis Paper [131]: The dataset consists of high-resolution WSIs of breast tissue specimens stained using IHC and H&E techniques. Sourced from Bahcesehir University Medical School, the dataset is structured into two subsets: BAU_IHC, containing 163 slides across 55 zip files, and BAU_HE, comprising 72 slides across 36 zip files, with files ranging from a few hundred megabytes (MB) to over a gigabyte (GB) in size. These images, provided in .svs format, serve as a valuable resource for histopathology research, deep learning-based tissue segmentation, and computational pathology applications. The dataset is accompanied by a GitHub repository featuring Python-based tissue region segmentation code, which enables automated analysis of WSIs for potential diagnostic insights. Approved by the Bahcesehir University Clinical Research Institutional Review Board.
119. TCGA-UT - Histology Images from Uniform Tumor Regions in TCGA WSIs [132]: Originating from TCGA, this dataset spans 32 cancer types, totaling 8736 WSIs and 271,000 high-quality patches. Patches are resized to 256 $\times$ 256 pixels, retaining details crucial for histopathological studies. Annotated for uniform tumor regions, this dataset facilitates deep learning tasks such as texture representation extraction, CBIR, and genomic feature prediction, with quality oversight from trained pathologists.
120. CRAG [133]: Used for gland instance segmentation in colorectal cancer, the CRAG dataset provides 173 training and 40 validation images with 1500 $\times$ 1500 pixel patches at 20$\times$ magnification. It captures diverse glandular morphologies, supporting segmentation tasks in colorectal cancer diagnosis. Annotations highlight well to poorly differentiated structures, assisting in developing accurate gland segmentation models, as demonstrated in frameworks like MILD-Net.
121. PESO [134]: The PESO, prostate epithelium segmentation on H&E-stained prostatectomy WSIs dataset provides WSIs of prostatectomy specimens for deep learning-based epithelial segmentation in prostate cancer (PCa). It includes 62 training WSIs and 40 test WSIs, exported at 0.48 μm/pixel resolution. The training set contains segmentation masks generated using a U-Net trained on IHC-stained slides, along with color deconvolution masks highlighting P63 and CK8/18-stained regions, with 25 manually corrected masks to remove artifacts. The test set includes annotated test regions (2500 $\times$ 2500 pixels per region), ground truth segmentation masks, and cancer/benign labels in a CSV file.
122. TCGA [135]: A comprehensive NIH initiative that covers more than 33 cancer types, TCGA provides a wealth of genomic, transcriptomic, proteomic, and clinical data. The dataset integrates molecular details with histopathological and clinical annotations, allowing researchers to study the relationships between genetic mutations, tissue morphology, and clinical outcomes. The breast cancer subset, with over 1200 cases, includes comprehensive molecular profiles, supporting studies in cancer biology, potential therapeutic targets, and biomarker identification across various cancers.
123. HunCRC [136]: A colorectal cancer dataset with 200 WSIs from H&E-stained biopsy samples, scanned at 40$\times$ magnification (0.1213 μm/pixel). It includes over 101,389 patches (512 $\times$ 512 pixels) annotated for ten pathological categories such as low-grade dysplasia, adenocarcinoma, and tumor necrosis. Local and global annotations, validated by a certified pathologist, ensure data quality, supporting fine-grained pathological analysis and model development in colorectal cancer screening.
124. TissueNet: Created for the Detect Lesions in Cervical Biopsies challenge, TissueNet contains high-resolution WSIs of cervical biopsies from French medical centers, totaling 700 GB. Each slide, up to 150,000 $\times$ 85,000 pixels, is classified into four lesion categories: benign, low malignant potential, high malignant potential, and invasive cancers. This large-scale, high-resolution dataset supports model development for cervical cancer detection and requires efficient computational approaches due to its data volume.
125. AML-Cytomorphology_LMU [137]: This acute myeloid leukemia (AML) dataset includes over 18,000 annotated images of white blood cells from 200 patients, captured at 100$\times$ magnification with Wright's stain. Each image

is labeled to identify malignant and non-malignant cells, aiding deep learning models in accurately detecting blast cells. Publicly available on TCIA, it serves as a benchmark for automated AML diagnosis and clinical decision support, promoting advancements in blood smear analysis.

126. CPTAC-AML [138]: This dataset from the CPTAC includes 122 high-resolution histopathology images from 88 AML patients, in SVS format, along with extensive clinical metadata. The dataset supports both discovery and targeted proteomic studies, allowing for comprehensive protein analysis related to AML. With demographic, classification, and follow-up data, this dataset is ideal for machine learning applications that explore AML's proteogenomic characteristics. It is available on TCIA under a CC BY 3.0 license, accessible via the IBM-Aspera-Connect plugin.
127. CRC-TP [139]: Designed for colorectal cancer research, the CRC-TP dataset includes 280,000 patches from 20 WSIs, annotated for seven tissue phenotypes such as tumor, benign, and necrotic tissue. Additionally, the CRC-CC subset offers cellular-level classification for five cell types, providing a detailed resource for colorectal cancer tissue segmentation and phenotyping. This dataset supports advancements in computational pathology, facilitating automated tissue characterization and phenotypic analysis.
128. DLBCL-Morph [140]: Focused on Diffuse Large B-Cell Lymphoma (DLBCL), this dataset contains tissue microarray images of 209 cases with H&E and IHC staining (e.g., CD10, BCL6). Annotated by pathologists, the dataset is used for nuclei segmentation and morphological analysis, employing the HoVer-Net model to capture geometric features valuable for prognosis. Survival data allows Cox modeling, achieving a C-index of 0.635 based solely on geometric features. Available through the Stanford ML Group GitHub, it aids research in DLBCL morphology and survival prediction.
129. A Dataset and a Technique for Generalized Nuclear Segmentation for Computational Pathology [141]: This dataset provides 30 H&E-stained tissue images from multiple organs and disease states, featuring over 21,000 nuclei with detailed boundaries. Validated by expert pathologists, it includes diverse samples to train machine learning models in nuclear segmentation across various nuclear appearances, including challenging densely packed and chromatin-sparse nuclei. It serves as a robust benchmark for advanced segmentation techniques, supporting models in generalizing well to new datasets.
130. SegPC-2021 [142, 143]: Targeted for plasma cell segmentation in multiple myeloma (MM) diagnostics, this dataset consists of 775 bone marrow aspirate images from AIIMS, New Delhi, India, stained with Jenner-Giemsa. Images are captured at two resolutions and color-normalized, addressing segmentation challenges like cell clustering and background interference. Annotations differentiate nucleus and cytoplasm, offering a detailed training set for machine learning. SegPC-2021 was featured in IEEE ISBI 2021 for evaluating segmentation performance on an active leaderboard.
131. UMMC ER-IHC Breast Histopathology Whole Slide Image and Allred Score [144]: This dataset from Universiti Malaya Medical Centre, Malaysia, includes 37 ER-IHC WSIs for breast cancer, scanned at 20× magnification, with Allred scores verified by pathologists to distinguish ER-positive and ER-negative cases. Pathologist-defined ROIs in WKT format support computational scoring, with CSV data on ROI characteristics. Nuclei segmentation counts were obtained with a Stardist-HE model, facilitating automated ER status assessment and Allred scoring in breast cancer.
132. Benign Breast Tumor Dataset [145]: The Benign Breast Tumor Dataset supports AI applications in benign breast tumor analysis. It includes 80 high-resolution mammogram images and detailed patient data from 83 Indian patients, stored in a zip file and excel sheet. This dataset contains tumor subclassifications, mammographic findings, and histopathological characteristics, facilitating analysis for diagnostic tools to differentiate benign from malignant lesions. Originally developed for the AI Against Cancer Data Science Hackathon, it serves research in computational pathology and radiology by offering diverse imaging and clinical data.
133. Cytological and Histological Images of Breast Cancer [146]: This dataset from Ukrainian institutions provides cytological images at 3264 × 2448 px (40× lens) and histological images at 2048 × 1536 px, including IHC-stained images for four biomarkers: ER, PR, KI67, and HER2NEU. It supports breast cancer biomarker and image analysis research.
134. Reinterpretation of Prostate Cancer Pathology by Appl1, Sortilin, and Syndecan-1 Biomarkers [147]: This dataset provides a comprehensive digital pathology atlas for prostate cancer diagnosis. It includes H&E-stained tissue sections alongside three novel biomarker-stained sections (Appl1, Sortilin, and Syndecan-1), improving accuracy in cancer grading. The dataset covers benign cases and ISUP grades 1–5, enabling pathologists to refine diagnoses using biomarker-assisted histopathology. High-resolution images were acquired using a Carl ZEISS AxioScan.Z1 microscope at 40× magnification, stored in BigTIFF format for lossless compression, ensuring quality preservation.

135. 2 Million Histological Images of Breast Cancer Tumors with HER2 Labels [148, 149]: Sourced from the A.C. Camargo Cancer Center, Brazil, this dataset comprises 2 million H&E-stained patches (256 × 256 px at 0.5 μm/pixel) with HER2 status (HER2-negative, HER2-low, HER2-high) from 504 breast cancer patients. Captured at 40× magnification, it excludes cases of metastasis or pre-treatment, focusing on tumor regions. The dataset, available under CC BY 4.0 on Zenodo, is preprocessed with a ConvNext-tiny network and includes code for model training, aiding in HER2-based breast cancer analysis.
136. Computational Pathology to Discriminate Benign from Malignant Intraductal Proliferations of the Breast [150]: Focused on differentiating between ductal carcinoma in situ (DCIS) and usual ductal hyperplasia (UDH), this dataset includes 392 features (nuclear size, shape, intensity, texture) extracted from 116 breast biopsies at Massachusetts General Hospital. Using L1-regularized logistic regression, the model achieved a high area under the curve (AUC) in cross-validation (0.95) and external validation (0.86), highlighting predictive morphological features. The dataset, available on Dryad, also enables DCIS nuclear grade stratification with an AUC of 0.98, supporting breast lesion diagnostics in computational pathology.
137. Biological Classification of Breast Cancer by Real-Time Quantitative RT-PCR [151]: Designed to streamline breast cancer classification, this dataset validates a qRT-PCR-based assay using a 53-gene panel to classify tumors into Luminal, Normal-like, HER2+/ER−, and Basal-like subtypes. Tested on 123 samples, the qRT-PCR method demonstrated 91% concordance with microarray classifications. Additionally, Luminal tumors correlated with better outcomes, while certain genes predicted relapse-free survival independently of grade and stage, showcasing qRT-PCR's potential in prognostic breast cancer subtyping.
138. The Single-Cell Pathology Landscape of Breast Cancer [152]: This imaging mass cytometry (IMC) dataset includes 720 pathology images from 352 breast cancer patients, using a 35-antibody panel to capture cellular heterogeneity and spatial organization. Covering ER, PR, HER2, Ki-67, and more, it enables single-cell insights into tumor microenvironments and identifies unique breast cancer subgroups with distinct spatial patterns that correlate with clinical outcomes. This dataset provides a granular view of cell interactions within tumors, enhancing breast cancer classification at a single-cell level for potential personalized diagnostics.
139. Type and Duration of Exogenous Hormone Use Affects Breast Cancer Histology [153]: This dataset, based on a 2003 cohort from the Kaiser Permanente Northern California Cancer Registry, includes 2830 breast cancer cases. It investigates the impact of hormone replacement therapy (HRT) type and duration on cancer histology, using pharmacy records to classify HRT as none, estrogen-only, progesterone, or combined. Results indicate that prolonged combined HRT (over six months) increases the likelihood of ER-positive tumors, while estrogen-only HRT is associated with lower-grade, earlier-stage cancers. The findings emphasize the duration-dependent effects of HRT on breast cancer risk and phenotype.
140. Automated Nuclear Pleomorphism Scoring in Breast Cancer [154, 155]: Comprising 118 WSIs of breast cancer tissue stained with H&E, this dataset from Radboud University Medical Center aids in scoring nuclear pleomorphism—a marker of cancer aggressiveness. Originally scanned at 0.25 μm/px, images are available in TIFF format at 0.5 μm/px for accessibility. Serving as a benchmark for automated pleomorphism scoring algorithms, this dataset is accessible on Zenodo, supporting studies that assess tumor cell variation in breast cancer.
141. Unnormalized Breast Cancer Histopathology: Created by Sourabh Kumar, this dataset includes 2480 histopathology images divided into benign (1240) and malignant (1240) categories. Released in November 2022, it provides an unnormalized set of breast cancer images, useful for training and evaluating machine learning models that classify breast cancer without prior normalization. This straightforward organization supports early-stage model development for benign versus malignant tissue classification.
142. Multi-omic Machine Learning Predictor of Breast Cancer Therapy Response [156]: This dataset includes multi-omic profiles (clinical, genomic, transcriptomic, and digital pathology) of breast cancer biopsies from 168 patients, aimed at predicting therapy response. Data comprises DNA and RNA sequencing and digital pathology with assessments of immune infiltration and mutation burden. Key features such as TP53 and PIK3CA mutations and immune signatures correlate with therapy outcomes like residual disease and complete response, validated on an independent cohort of 75 patients. The dataset supports predictive modeling of response to chemotherapy, HER2-targeted therapy, and neoadjuvant treatments.
143. CBIS-DDSM [157]: Released in 2017, the curated breast imaging subset of digital database for screening mammography (CBIS-DDSM) dataset offers 3103 mammogram images with over 3500 labeled abnormalities, sourced from institutions like Massachusetts General Hospital and Wake Forest University. It includes

detailed annotations, such as mass shapes, margins, BI-RADS categories, and pathology statuses (benign or malignant), across craniocaudal (CC) and mediolateral oblique (MLO) views. With pixel-level instance segmentation and unified train-test splits, this dataset is ideal for advancing AI in mammography-based detection and diagnosis.

144. Lung Adenocarcinoma Evolution H&E Pathomic Feature Analysis Dataset [158]: Focused on lung adenocarcinoma evolution, this 2023 dataset includes H&E-stained histopathology images from 98 patients across Japan, China, and the USA. It covers 162 slides and 669 ROIs across various histological stages, from normal regions to advanced adenocarcinoma, with robust cell segmentation and nine pathomic features. Structured data files support analysis of cell classification, clinical demographics, and histological progression, making it a valuable resource for studying lung cancer diagnostics and evolutionary progression.
145. Fused Radiology-Pathology Lung Dataset [159, 160]: Curated by Case Western Reserve University, USA, this dataset integrates radiology (computed tomography (CT)) and pathology data to analyze invasive adenocarcinoma in small ground-glass pulmonary nodules. Covering six patients with 11,210 CT images (in DICOM format) and annotated histopathology (in TIF format), it offers a precise imaging ground truth by excluding larger nodules. Clinical data in Excel format is included, and analysis scripts are available on GitHub. This dataset is ideal for validating diagnostic imaging models for adenocarcinoma.
146. Colsanitas Dataset [161, 162]: Originating from the Colsanitas Clinic in Bogota, Colombia, this dataset includes 544 WSIs from 80 breast cancer patients, digitized at 40× magnification from H&E-stained slides. Expert pathologists annotated 2250 images across four classes: normal tissue, benign lesions, in situ carcinoma, and invasive carcinoma. Saved in TIF format, the dataset supports multi-class classification, with stain normalization applied to account for color variability, making it a valuable resource for developing breast cancer diagnostic models.
147. CPM-17 [163]: Derived from TCGA, this dataset includes high-resolution WSIs of non-small cell lung cancer (NSCLC), glioblastoma multiforme (GBM), lower grade glioma (LGG), and head and neck squamous cell carcinoma (HNSCC). It provides 32 manually segmented image tiles per cancer type for nuclei segmentation and diagnostic regions for NSCLC classification. Capturing cellular structures in detail, it is well-suited for both fine-grained segmentation and broader diagnostic classification tasks in cancer research.
148. CPM-15 [163]: Focused on brain tissue samples, this dataset includes 15 H&E-stained image tiles at 20× and 40× magnifications, stored on Google Drive for accessibility. With a total of 2905 segmented nuclei, the images range in size (400 × 400 and 600 × 1000 pixels), allowing both wide-view and detailed analysis of brain tissue. It is well-suited for advanced segmentation tasks, providing cellular-level details for segmentation research.
149. International Cancer Genome Consortium (ICGC) Breast Cancer Histology Images: This dataset, hosted on the European Genome-Phenome Archive (EGA), includes 151 H&E-stained fresh frozen breast cancer tissue samples collected by the ICGC with the BASIS project. Each sample is linked to whole genome sequencing data from a foundational 2016 study by Nik-Zainal et al. [164] offering a rich combination of genetic and histological data. The dataset supports in-depth research into breast cancer biology, tumor heterogeneity, and progression, facilitating advancements in personalized medicine and cancer research.
150. CRC-ICM [165]: The CRC-ICM dataset is a specialized collection of IHC images from 136 colorectal cancer patients, aimed at analyzing immune cell activity within the tumor microenvironment (TME). It includes 1756 high-resolution RGB images (in JPEG format) capturing immune markers in both the center of the tumor and invasive margin regions. The dataset highlights six key immune markers—CD3, CD8, CD45RO, PD-1, Tim3, and LAG3—providing insights into immune infiltration and immune checkpoint activity. Each image is accompanied by clinical and pathological data, including tumor location, TNM stage, and differentiation grade, enabling in-depth analysis of immune behavior across different tumor regions.
151. SurGen [166]: The SurGen dataset contains WSIs coupled with clinical and genetic data concentrating on the colorectal regions along with their metastatic sites. It contains 10,202 WSIs from 843 cases with survival data available for 426 cases for enabling analysis of WSIs along with the survival genetic markers. Detailed annotations for key genetic mutations (KRAS, NRAS, BRAF) as well as data points on tumor site, patient age, sex, survival, and MMR/MSI status are provided that can help predict mismatch repair status from the WSIs for colorectal cancer.

Based on the above review of the pathology datasets, we observed the following trends. In Table 1, we categorized each dataset based on the primary organ or tissue type described. Datasets covering a wide range of organs were placed in a “Multi-organ” category. The data clearly shows

Table 1 Distribution of the 151 reviewed datasets by primary organ or tissue type

Organ/tissue type	Number of datasets
Breast	43
Colorectal	19
Prostate	10
Lung	9
Multi-organ/general	9
Brain/nerve	8
Gastric/stomach	6
Skin	5
Bone marrow/blood	5
Liver	5
Kidney	4
Oral	4
Ovarian	3
Endometrial	2
Bladder	2
Other	8
Total	151

a heavy focus on breast cancer, which accounts for over a quarter of all specialized datasets. This is followed by colorectal and prostate cancer, highlighting the areas where AI in pathology has seen the most research and data generation. The "Other" category includes single datasets for esophagus, pancreas, cervix, etc. We categorized the reviewed datasets in Table 2 based on the main task they were designed for, such as segmenting nuclei or classifying tissue types. Many datasets support multiple tasks; this table reflects the most prominent primary task mentioned. Segmentation is the most common task, appearing in nearly half of the datasets. This is driven by the fundamental need to identify structures like nuclei, cells, and glands before higher-level analysis can be performed. Classification of tissue patches (e.g., benign vs. malignant) is the second most common task. Tasks like "Detection" often focus on specific objects like mitotic figures or metastases,

Table 2 Distribution of the 151 reviewed datasets by the primary artificial intelligence (AI) task they are designed to support

Primary AI task	Number of datasets
Segmentation	69
Classification	54
Detection	12
Multi-task/general purpose	9
Other (registration, VQA, etc.)	7
Total	151

Table 3 Timeline of dataset publication years based on the primary citation for each of the 151 datasets. A significant increase in publications is notable from 2019 onwards

Year of publication	Number of datasets
2006–2014	4
2015	2
2016	6
2017	4
2018	6
2019	12
2020	17
2021	24
2022	26
2023	29
2024–2025	21
Total	151

while "Segmentation" is broader, covering nuclei, glands, and tumor regions. Finally, from Table 3, we notice that there is a dramatic acceleration in dataset publication since 2019, peaking in 2021–2023. This timeline directly reflects the explosion of interest and research in deep learning for digital pathology. While our systematic review identified 151 publicly available datasets, a subset of these have gained significant prominence, serving as common benchmarks for developing and validating new AI models. Many of these influential resources originated from grand challenges or large-scale research consortia, which have catalyzed innovation and established standards for performance evaluation in the field. To provide a focused overview of these key datasets, Table 4 summarizes 15 of the most popular and highly cited datasets. The table details each dataset's primary organ or tissue type, its main AI task, and its distinguishing features, such as scale or origin. This curated list includes foundational resources such as the multi-modal TCGA-BRCA collection, pioneering challenge datasets like CAMELYON16 for metastasis detection, and recent, large-scale benchmarks like the PANDA challenge for Gleason grading, which contains over 12,000 WSIs. Also highlighted are datasets crucial for fundamental tasks, such as MoNuSeg for multi-organ nuclei segmentation and NCT-CRC-HE-100K for colorectal tissue classification.

Discussion

The computational pathology domain has seen tremendous progress in the last decade due to the widespread application of AI models. Despite the vast improvement obtained by AI models compared to previous image processing and

Table 4 Summary of popular and highly-cited public datasets for computational pathology, selected from the full review. These datasets are frequently used for benchmarking AI models

Dataset name	Primary organ/tissue	Primary task(s)	Key features/scale
BreakHis	Breast	Classification	7909 images from 82 patients across 4 magnifications
CAMELYON16	Lymph node (breast)	Metastasis detection	Pioneering grand challenge dataset with H&E-stained WSIs
CAMELYON17	Lymph node (breast)	Metastasis classification	899 WSIs from 5 medical centers, focusing on TNM staging
TUPAC16	Breast	Proliferation scoring	821 WSIs with mitotic scores and gene expression data
MIDOG Series	Multi-organ (human/canine)	Mitotic figure detection	Multi-scanner, multi-species data designed for domain generalization
MoNuSeg	Multi-organ	Nuclei segmentation	H&E images from various organs; 22,000 nuclear annotations
GlaS	Colon	Gland segmentation	165 images of colorectal adenocarcinoma for segmentation challenges
PANDA	Prostate	Gleason grading	Large-scale challenge with 12,625 WSIs from multiple countries
NCT-CRC-HE-100K	Colorectal	Tissue classification	100,000 histological patches across 9 tissue classes
ICIAR 2018 (BACH)	Breast	Classification (4 classes)	400 microscopy images and WSIs for multi-class classification
PanNuke	19 Tissue types	Nuclei Seg and Class	Pan-cancer dataset with 200,000 nuclei across 5 cell classes
Lizard	Colon	Nuclei Seg and Class	Large-scale with nearly 500,000 annotated nuclei from colonic tissue
BCSS	Breast (TNBC)	Semantic segmentation	151 WSIs with over 20,000 annotated regions (tumor, stroma, etc.)
CRAG	Colorectal	Gland segmentation	213 images with diverse glandular morphologies for segmentation
TCGA-BRCA	Breast	General/multi-modal	3111 WSIs linked to comprehensive genomic and clinical data

machine learning approaches, the progress is dependent on large-scale pathology datasets that require careful acquisition and metadata aggregation. To the best of our knowledge, there are no reviews that focus on open pathology datasets whether they are academic or commercially available with details. Our systematic review identified and cataloged 151 publicly available datasets for AI in histopathology, revealing a rapidly growing but unevenly distributed landscape. The dramatic increase in dataset publications since 2019 reflects the concurrent explosion of interest in deep learning and AI for pathology. While the proliferation of data is a positive sign, our analysis highlights several key trends, gaps, and challenges that the research community must address to continue advancing the field.

Assessment of Dataset Quality Based on FAIR Principles

As the volume of histopathology data continues to grow, ensuring that datasets are machine-actionable and accessible with minimal human intervention is critical. We evaluated the included datasets based on the Findability, Accessibility, Interoperability, and Reusability (FAIR) Guiding Principles [167], identifying a notable variation in how data is curated and shared.

- Findability (F): Findability requires that (meta)data be assigned a globally unique and persistent identifier (PID) such as a Digital Object Identifier (DOI).
  - Strengths: Major challenge datasets like CAMELYON16 and TCGA-BRCA are indexed in searchable repositories like The Cancer Imaging Archive (TCIA), ensuring they are easily discoverable by the global research community.
  - Gaps: Our review found that the majority of the 151 datasets lack a standard DOI or a permanent link, often residing on personal or institutional websites that may suffer from "link rot."
- Accessibility (A): Accessibility focuses on the ability to retrieve data through standardized communication protocols.

  - Public vs. controlled: While 151 datasets were selected for their public availability, some (e.g., GTEx Portal) require authorized access to protect patient privacy.
  - Retrieval protocols: Top-tier datasets utilize open protocols for metadata retrieval, allowing researchers to verify the content (e.g., number of WSIs or magnification) before full download.

- Interoperability (I): Interoperability ensures that data can be integrated with other datasets or analysis workflows.
  - Standard formats: Higher-quality datasets utilize widely accepted file formats like BigTIFF, SVS, or DICOM for whole-slide images and JSON or XML for annotations.
  - Multimodal gaps: A significant barrier to interoperability remains the lack of shared vocabularies and ontologies across histopathological, radiological, and genomic data, making multimodal fusion difficult.

- Reusability (R): Reusability is the ultimate goal, requiring rich metadata and clear usage licenses.
  - Provenance and licensing: Datasets like MoNuSeg and PanNuke provide detailed provenance (staining protocols, organ types) and are released under clear licenses (e.g., CC BY-NC-SA 4.0), facilitating their use as benchmarks for new AI models.
  - Benchmarking: Standardizing these resources allows for "horizontal evaluation," where different AI models are tested against the same standardized data to accurately compare performance.

To rigorously evaluate the usability and long-term sustainability of the identified research resources, a systematic assessment of the FAIR principles was conducted for all 151 datasets. This evaluation utilized the F-UJI automated assessment framework, a recognized tool for programmatically measuring research data "FAIRness" based on 16 core metrics [168]. By assessing the machine-readability of metadata and the persistence of identifiers (e.g., DOIs) for each dataset landing page, we quantified the level of data stewardship currently present in the field of computational pathology. The full assessment is provided in the supplementary material and highlights a critical need for standardized data practices. Our analysis indicates that while data volume is rapidly expanding, a significant portion of publicly available resources lacks the persistent identifiers and standardized metadata required for seamless cross-institutional integration. Adopting the FAIR principles is therefore not merely a technical requirement but a clinical necessity to ensure that AI-ready data is truly accessible, interoperable, and reproducible for the global development of diagnostic tools.

The analysis of 151 datasets reveals a mean total FAIR score of 55.6%, with a notable clustering of "high-performers" reaching scores as high as 92%. The top-tier datasets, including HistMNIST, TNBC, Naylor et al., and the Fused Radiology-Pathology Lung Dataset, consistently leveraged established repositories such as Zenodo, TCIA, and Dryad. These platforms ensure high scores in Findability (mean 4.15/7) and Accessibility (mean 4.01/7) by providing persistent DOIs and standardized retrieval protocols. A significant finding from this assessment is the discrepancy between data volume and data stewardship. While the field has seen an explosion in the number of images, Interoperability (mean 3.13/6) and Reusability (mean 3.31/6) remain the primary bottlenecks. Many datasets hosted on institutional websites or personal cloud storage lacked the machine-readable metadata and clear licensing (e.g., Creative Commons) necessary for automated cross-institutional workflows. The lack of high scores in Interoperability suggests that a "manual overhead" still exists for researchers attempting to fuse disparate pathology datasets. For clinical AI to move toward foundation models which require the ingestion of massive, varied data, the community must transition away from isolated web links toward standardized data objects. The quantitative scores provided in the supplementary material serve as a roadmap for this transition, identifying which resources are truly "AI-ready" and which require better curation to meet the standards of reproducible medical science. The clinical significance of these datasets is clear: they provide the ground truth necessary to reduce diagnostic subjectivity and automate labor-intensive tasks like metastasis detection and biomarker scoring. Yet, for these AI tools to be truly "clinical-grade," the community must move toward FAIR principles ensuring that data is not only available but also findable, interoperable, and reusable across diverse institutional settings.

Horizontal Evaluation of Dataset Applicability

A key challenge in computational pathology is that performance on one dataset rarely translates directly to another due to "domain shift," variability caused by different scanners, staining protocols, and tissue types. To help researchers select the most appropriate resources, we evaluate the landscape of datasets for the two most common tasks: nuclei segmentation and tissue classification.

1. Nuclei Segmentation - From Specialized to General-Purpose: While multiple popular datasets support nuclei segmentation, they vary significantly in their "difficulty" and intended use case. MoNuSeg (Generalist Benchmark) offers a high-diversity, multi-organ approach (lung, prostate, kidney, breast) with 22,000 nuclear boundary annotations. It is the gold standard for testing a model's generalizability across different human organs

and staining protocols. PanNuke (Multi-Class Granularity) provides nearly 200,000 segmented nuclei across 19 tissue types. Unlike MoNuSeg, it categorizes nuclei into five cell classes (e.g., neoplastic, inflammatory, dead cells), making it better suited for tumor microenvironment (TME) analysis rather than simple detection. Lizard (Large-Scale Colonic Focus) contains nearly 500,000 annotated nuclei specifically from colonic tissue. This is ideal for researchers focusing on colorectal cancer profiling, where the massive volume of data allows for training deep learning models that are robust to clinical "noise."

2. Tissue Classification - Binary vs. Multi-Class Complexity: Classification tasks range from simple "benign vs. malignant" binary decisions to complex multi-tissue staging. BreakHis (Binary Baseline) focuses on breast cancer images across four magnifications (40×–400×). Its clear-cut binary and subtype labels make it an excellent entry point for testing new architectures on a stable, well-understood dataset. NCT-CRC-HE-100K (Multi-Tissue Categorization) offers 100,000 patches across nine distinct tissue classes (e.g., stroma, muscle, debris, tumor). This dataset is far more challenging than BreakHis as it requires the model to differentiate between various non-cancerous structures, which is essential for developing real-world automated diagnostic tools.

When performing a horizontal evaluation, researchers must account for several technical variables that influence performance, (1) magnification and resolution—models trained on 20× datasets (e.g., MoNuSAC 2020) may fail on 40× datasets (e.g., Janowczyk et al [35]) because of differences in visible cellular detail; (2) annotation granularity—there is a significant performance gap between models trained on weakly supervised image-level labels (e.g., CAMEL) and those trained on pixel-level ground truth masks (e.g., PAIP 2023); and (3) stain normalization—some datasets, like NCT-CRC-HE-100K-NONORM, explicitly exclude color normalization to force models to learn "stain-agnostic" features, which is significantly more difficult but leads to more robust clinical tools.

Generative AI and the Future of Histopathology Data

Current histopathology FMs, such as Virchow, UNI, and GIGAPATH, are trained on datasets ranging from 100,000 to over 1.5 million WSIs, encompassing billions of image tiles. These models create high-dimensional feature representations that can be "fine-tuned" for specific clinical tasks using a fraction of the labeled data required by previous architectures. This shift underscores the clinical significance of large-scale repositories like TCGA and PANDA, which provide the sheer volume of data necessary to train these "knowledge-rich" backbones. A major gap identified in our review is the scarcity of datasets for rare malignancies and under-represented patient groups. Generative models, particularly generative adversarial networks (GANs) [169] and diffusion models [170], offer a potential solution. For data augmentation, researchers can now generate high-fidelity synthetic histopathology patches to balance datasets where certain classes (e.g., rare tumor subtypes or specific genetic mutations) are under-represented. Along with augmentation, expert-guided image generation can elevate AI diagnostic accuracy by creating diverse, realistic edge cases for model training, helping to bridge the gap in clinical "noisy" data. Generative AI also holds promise in synthesizing one modality from another, for example, generating IHC-like staining patterns from H&E-stained slides, as explored in datasets like HEROHE and ACROBAT 2023. Further, while multimodal datasets are currently scarce, generative AI, large language models (LLMs), and vision-language models (VLMs) provide new ways to integrate disparate data types [171]. Datasets such as PathVQA (question-answering) and ARCH (multiple instance captioning) are foundational for training models that can interpret both pathology images and clinical text, moving toward a more holistic, clinician-like "reasoning" process.

Challenges and Future Directions

Based on our review, we found the following points along with some challenges that still remain in creating and maintaining high-impact large-scale pathology datasets:

- Dominance of segmentation and classification tasks: Public, well-curated, well-cited datasets are prevalent in the nuclei, cell segmentation field whereas a relatively small number of datasets tackle other pathology challenges such as oncology classification and biomarkers discovery [172]. Further, even within the realm of nuclei and cell segmentation datasets, the quality of annotations provided per dataset differ highly, and to develop AI models that can work across datasets require careful attention must be paid when different datasets as some of them have data overlap.
- The dataset landscape is dominated by a few key cancer types and AI tasks: Our analysis shows a heavy concentration of resources for breast, colorectal, and prostate cancers. This focus, while productive, leaves many other malignancies under-resourced. Furthermore, the most common tasks addressed are segmentation and classification. Foundational segmentation datasets like MoNuSeg, GlaS, and PanNuke have been crucial for developing models that can identify basic primitives like nuclei and glands. However, this focus on fundamental tasks means fewer datasets are available for more

complex clinical challenges, such as predicting treatment response or patient survival.

- The challenge of heterogeneity and generalization: Though the Cancer Genome Atlas (TCGA) has a sizable H&E datasets and other associated omics for various cancers, their lack of WSI-specific region of interests (ROIs) and their corresponding labels limit the clinical applicability of developed AI models since generalization is difficult.
- Dearth of truly big data: There are several general pathology H&E datasets, for example, MedMNIST [173, 174] contains PathMNIST as a subset and is useful in developing proof-of-concept pathology-specific AI models. However, they are still not large enough to develop robust pathology AI models that can be used across multiple tasks. For example, PathMNIST presents a 9-class classification for AI models and contains 100,000 patches for training 7180 for testing.
- The gap in multimodal datasets: Despite an overall increase in the availability of datasets for pathology, multimodal datasets are not as prevalent—cancer and many other diseases require a holistic approach of utilizing multiple data sources—imaging, genomics, patient records, and clinical data [175–178]. A significant gap identified in our review is the scarcity of multimodal datasets. Cancer is a complex disease driven by interconnected factors at the genomic, proteomic, and tissue levels. While a few notable datasets, such as the CPTAC collections and UPENN-GBM, integrate histopathology with omics or radiological data, they are the exception rather than the rule. The future of precision medicine depends on AI models that can synthesize these disparate data types. Advancing the field will require a concerted effort to create more datasets that link whole-slide images to corresponding genomic, transcriptomic, and clinical outcomes.
- Geographic bias and regional disparities: Our analysis reveals a significant geographic concentration of data. A majority of the high-impact, large-scale datasets (e.g., TCGA, CAMELYON, PANDA) originate from medical centers in North America and Europe (Western Centricity). There is a critical dearth of datasets from the Global South, including Africa, Latin America, and parts of Southeast Asia, which may result in "population shift" where models fail to account for genetic and environmental variations specific to these regions (under-represented populations). For example, demographic bias can lead to marked performance disparities in subtype breast (TCGA-BRCA) and lung carcinomas (TCGA-lung cohorts) and IDH1 mutation prediction in brain tumor (TCGA-GBMLGG cohort [5]) prediction as shown by [179]. Also, AI models trained exclusively on high-resource institutional data may not be equitable or perform well in low-resource settings where staining protocols or scanning equipment differ significantly (Healthcare Inequity). Unlike in radiology AI [180] which is a mature field now, in the pathology AI area, understanding biases and disparities and how to overcome them remains relatively under-explored [181–184].
- Annotation quality and inter-observer variability: The ground truth of any pathology dataset is only as reliable as the specialists who created it. Inconsistent standards: We observed high variability in annotation quality across the reviewed datasets, ranging from pixel-level masks validated by multiple pathologists to weakly-labeled images where "ground truth" is inferred from clinical reports. Subjectivity: Even expert pathologists exhibit inter-observer variability in complex tasks like Gleason grading or mitotic counting. This subjectivity is baked into the training data, potentially limiting the absolute accuracy an AI model can achieve. Crowdsourcing trade-offs: Some large-scale datasets, such as NuCLS, utilize crowdsourced labels. While this enables massive data scaling, it introduces "noisy" labels that require sophisticated algorithmic handling to ensure clinical reliability.
- Technical and interpretive gaps: task distribution: The dataset landscape is dominated by segmentation and classification, leaving complex clinical challenges like survival prediction or longitudinal disease progression under-resourced. Multimodal scarcity: Despite the clinical necessity of integrating genomics and radiology, the review identified a persistent gap in datasets that link WSIs with other data modalities, limiting the development of truly "holistic" AI models. Snapshot in time: Due to the exceptional rate of growth in digital pathology, it is highly likely that new datasets have been released since the conclusion of our search in late 2025.
- A call for standardization in dataset citation: Finally, through our systematic review, we identified a challenge in properly citing the datasets as research artifacts remain inconsistent. A persistent challenge is the lack of standardization in how datasets are curated and cited. Recently, there have been calls to standardize and recognize the efforts in organizing scientific datasets [185] and using DOI as a permalink for retrieving the datasets. Despite this, in our review we found that many of the 151 datasets we covered here do not have neither a standard DOI nor a standard citation that can be attributed to the datasets' authors. Promoting a community-wide standard for data publication and citation is essential for building a more robust and reproducible research ecosystem. The presence of datasets with lower than 30 and near-zero FAIR scores underscores the risk of "dark data"—valuable clinical resources that remain effectively invisible to automated search and retrieval systems.

Based on our findings, we propose several future directions for the community:

- Diversify data collection: There is a critical need for more datasets focused on rare cancers and under-represented patient populations to ensure that AI models are equitable and generalizable.
- Prioritize multimodal datasets: Future data collection efforts should prioritize linking histopathology images with other data modalities, including genomics, proteomics, and radiology, to enable more holistic disease modeling.
- Develop a centralized portal: The creation of a centralized, community-maintained portal for tracking, accessing, and citing pathology datasets would be an invaluable resource. This would improve visibility, reduce redundancy, and facilitate collaboration.
- Standardize annotation and curation: Establishing community-wide best practices for annotation quality, metadata reporting, and data formatting would significantly enhance the utility and reusability of public datasets.

Finally, in terms of our systematic review, we identified some limitations. First, despite our comprehensive search strategy across multiple databases, we may have missed datasets that are not formally indexed, are published on institutional websites, or are in non-English journals. Second, the categorization of datasets by primary task and organ is based on the descriptions provided in the source publications and involves a degree of interpretation. Finally, the field is advancing at an exceptional rate, and new datasets have likely been released since our search was concluded.

Conclusion

Computational pathology has made strong strides in the diagnosis of multiple diseases including but not limited to cancers. Artificial intelligence (AI) techniques are now increasingly utilized in the histopathology domain to infer and provide computer-aided diagnosis (CAD) solutions that can help improve the decision-making. Availability of large-scale datasets is crucial in developing AI-driven solutions for histopathological challenges, and in this review, we provided details of publicly available datasets that are in various tissue types, organs, and cancers. By categorizing the datasets in terms of patients, organs, staining, and magnification factors, we provided a listing that can be useful in computational pathology research. Despite the recent proliferation of public datasets, recent advances in foundation models in computational pathology require millions of well-curated whole-slide images (WSIs), and further multimodal datasets are needed to tackle complex human health-related problems. Addressing the current limitations and gaps in the availability of such datasets requires continued collaboration and investment from researchers, healthcare institutions, regulatory bodies, and technology providers. Efforts must focus on improving data quality, establishing standardized annotation protocols, developing robust data sharing platforms, and ensuring the inclusion of diverse patient populations in dataset development. Looking ahead, the future of AI in pathology is inextricably linked to the availability of high-quality, standardized, and diverse datasets. These resources will be instrumental in driving the next wave of innovation in the field, leading to the development of AI models with broader applicability, greater accuracy, and the potential to significantly advance precision medicine and improve healthcare outcomes for patients worldwide.

Supplementary information

Excel file: Table1_List_of_Datasets_Sep2025_FAIRScores.xlsx
Word file: Table1_List_of_Datasets_Sep2025.docx

Appendix. Abbreviations

Abbreviations Table 5.

Table 5 Abbreviations

AI	Artificial intelligence
AML	Acute myeloid leukemia
ALN	Axillary lymph node
AUC	Area under the curve
CAD	Computer-aided diagnosis
CBIR	Content-based image retrieval
CMC	Canine mammary carcinoma
CNN	Convolutional neural network
CPTAC	Clinical Proteomic Tumor Analysis Consortium
CRC	Colorectal cancer
CSV	Comma-separated values
DCIS	Ductal carcinoma in situ (DCIS)
DLBCL	Diffuse large B-cell lymphoma
DOI	Digital object identifier
eQTL	Expression quantitative trait loci
ER+	Estrogen receptor-positive
ET	Essential thrombocythemia
FFPE	Formalin-fixed paraffin-embedded
FM	Foundation model
FOV	Field of view

Table 5 continued

GB	Gigabytes
GBM	Glioblastoma multiforme
H & E	Hematoxylin and eosin
HCC	Hepatocellular carcinoma
HNSCC	Head and neck squamous cell carcinoma
HRT	Hormone replacement therapy
HSI	Hyperspectral images
IDC	Invasive ductal carcinoma
IHC	Immunohistochemistry
IMC	Imaging mass cytometry
ISH	In situ hybridization
ITC	Isolated tumor cells
JPEG	Joint Photographic Experts Group
JSON	JavaScript object notation
KNN	K-nearest neighbor
LGG	Lower grade glioma
LLM	Large language model
LUAD	Lung adenocarcinoma
MB	Megabytes
MF	Primary myelofibrosis
MM	Multiple myeloma
MPN	Myeloproliferative neoplasm
MSI	Microsatellite instability
NAT	Neoadjuvant chemotherapy
NSCLC	Non-small cell lung cancer
OSCC	Oral squamous cell carcinoma
PNI	Perineural invasion
PRISMA	Preferred reporting items for systematic reviews and meta-analyses
PV	Polycythemia vera
RCBi	Residual cancer burden index
RCC	Renal cell carcinoma
ROI	Region of interest
SCC	Squamous cell carcinoma
SEER	Surveillance, epidemiology, and end results
SVM	Support vector machine
TB	Terabytes
TCGA	The Cancer Genome Atlas
TCIA	The Cancer Imaging Archive
TIL	Tumor-infiltrating lymphocytes
TMAs	Tissue microarrays
TME	Tumor microenvironment
TNBC	Triple-negative breast cancer
UCC	Urothelial cell carcinoma
UDH	Usual ductal hyperplasia
VAE	Variational autoencoder
VLM	Vision-language model
VQA	Visual question answering
WBC	White blood cell
WHO	World Health Organization
WSI	Whole slide images
XML	Extensible markup language

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Author Contribution All authors discussed the results and contributed to the final manuscript that is submitted for consideration.

Availability of Data and Material The datasets covered by the current study are publicly available standard datasets. The details of the reviewed datasets and materials are provided in the Supplementary files.

Declarations

Consent to Participate Not applicable

Consent for Publication Not applicable

Conflict of Interest The authors declare no competing interests.

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